



Fast reactions and slow manifolds

Christian Kuehn and Jan-Eric Sulzbach

Abstract. In this paper we generalize the Fenichel theory for attracting critical/slow manifolds to fast-reaction systems in infinite dimensions. In particular, we generalize the theory of invariant manifolds for fast-slow partial differential equations in standard form to the case of fast reaction terms. We show that the solution of the fast-reaction system can be approximated by the corresponding slow flow of the limit system. Introducing an additional parameter that stems from a splitting in the slow variable space, we construct a family of slow manifolds and we prove that the slow manifolds are close to the critical manifold. Moreover, the semi-flow on the slow manifold converges to the semi-flow on the critical manifold. Finally, we apply these results to an example and show that the underlying assumptions can be verified in a straightforward way.

Mathematics Subject Classification. 35B25, 35A24, 37D10, 37L25.

Keywords. Geometric Singular Perturbation Theory, Slow Manifolds, Infinite Dimensions, Fast Reactions.

1. Introduction

The study of fast-reaction systems plays a crucial role in many natural sciences. Examples of fast-reaction systems and their limit can be found in reaction-diffusion models [22, 30, 49], enzyme reactions [26], coagulation-fragmentation models [5, 21] and many other bio-chemical systems [13, 27, 32]. The important aspect of fast-reaction limits is that they can help to reduce the complexity of a system by eliminating the fast evolving dynamics of the system and only the evolution of the slow dynamics remains. One approach, that stems from the modeling of chemical reactions, is to regard the fast-reaction limit as a quasi-steady-state approximation (QSSA) of the system [16, 25]. The mathematical study of fast-reactions in the context of reaction-diffusion dates back at least to the seminal work by Hilhorst et al. [33]. A more recent review of the topic can be found in [36]; we also refer to [6, 11, 12, 29] for several further examples analyzing fast-reaction systems.

One important question that arises in the study of fast-reaction systems is to rigorously prove the convergence of the system and its solutions towards the fast-reaction limit. Different approaches such as energy and entropy methods [3, 51], variational methods [15, 48, 53] or homogenization techniques [7, 47] have been proposed in recent years. Our approach follows a recent geometric viewpoint using dynamical systems methods as introduced in [35] for fast-slow systems. Here, the idea is to generalize the theory developed by Fenichel [23, 24] for ODE systems and combine it with techniques for infinite-dimensional Banach spaces cf. [9, 14, 31] and the references therein. The challenges and also several applications of this approach can be found in [17, 19, 35]. Yet, the case of fast-reaction system is not covered by the results in [35], which is the main focus of this work.

More precisely, the fast-reaction system analyzed in this work is the following semi-linear evolution equation

$$\begin{aligned}\partial_t u^\varepsilon &= Au^\varepsilon + \frac{1}{\varepsilon} f(u^\varepsilon, v^\varepsilon), \\ \partial_t v^\varepsilon &= Bv^\varepsilon + g(u^\varepsilon, v^\varepsilon),\end{aligned}\tag{1.1}$$

where $\varepsilon > 0$ is a small parameter, A and B are linear and densely defined operators on Banach spaces X and Y of scalar-valued functions respectively. Moreover, f and g are sufficiently regular nonlinear functions and $(u^\varepsilon(t), v^\varepsilon(t)) = (u^\varepsilon, v^\varepsilon) \in X \times Y$ are the unknown functions for some $t \in [0, \infty)$. We see that for $\varepsilon > 0$ the fast-reaction system (1.1) is equivalent to a fast-slow system of the form

$$\begin{aligned}\varepsilon \partial_t u^\varepsilon &= \varepsilon Au^\varepsilon + f(u^\varepsilon, v^\varepsilon), \\ \partial_t v^\varepsilon &= Bv^\varepsilon + g(u^\varepsilon, v^\varepsilon).\end{aligned}\tag{1.2}$$

The parameter ε gives rise to a formal time-scale separation between the fast variable u^ε and the slow variable v^ε . Letting $\varepsilon \rightarrow 0$ in either the fast-reaction equations (1.1) or the fast-slow system (1.2) yields the following slow subsystem

$$\begin{aligned}0 &= f(u^0, v^0), \\ \partial_t v^0 &= Bv^0 + g(u^0, v^0),\end{aligned}\tag{1.3}$$

which is an differential-algebraic equation defined on a critical set

$$S_0 := \{(u^0, v^0) \in X \times Y : f(u^0, v^0) = 0\},\tag{1.4}$$

We refer to S_0 as the critical manifold, where we shall later on make a precise technical assumption involving the implicit function theorem to ensure that it is indeed a Banach manifold.

The main result of our work is a generalization of Fenichel's theory for the fast-reaction system (1.1). The two key assumptions are that the critical manifold is attracting, which is encoded in suitable technical attraction conditions and that the operator B allows a splitting of the space Y into a fast and slow component. This splitting is denoted by a small parameter $\zeta > 0$, depending on ε . Under these assumptions we obtain several results that together give a generalization of Fenichel's theory to infinite dimensions for the case

of fast-reactions. The first result is that there exists a locally invariant and continuously differentiable manifold for the fast-reaction system (1.1) denoted by S_ε . Secondly, this slow manifold S_ε converges to the critical manifold S_0 as $\varepsilon \rightarrow 0$. Lastly, we show that the semi-flow on the slow manifold converges to the slow flow on a subset of the critical manifold S_0 . The precise technical statements of these results can be found below in Propositions 3.4-3.9. One key observation in the study of slow manifolds for fast-reaction systems of form (1.1) is that the linearization of the non-linearity f with respect to the fast variable u determines the behaviour of the fast component. To encompass this in our system we rewrite it as follows

$$\begin{aligned}\partial_t u^\varepsilon &= \frac{1}{\varepsilon}(\varepsilon A + D_u f(u, v))u^\varepsilon + \frac{1}{\varepsilon}\tilde{f}(u^\varepsilon, v^\varepsilon), \\ \partial_t v^\varepsilon &= Bv^\varepsilon + g(u^\varepsilon, v^\varepsilon),\end{aligned}\tag{1.5}$$

where the details are presented in Section 2.1. For this system we try to apply the results presented in [35]. However, many results cannot be applied one-to-one as one has to be careful working with the new linear operator $\tilde{A}_\varepsilon = \varepsilon A + D_u f(u^0, v^0)$, where $D_u f$ denotes the Fréchet derivative in the direction of the first variable evaluated at (u^0, v^0) .

Now, let us briefly outline the structure of this work. We show that the fast-reaction systems are well-posed and prove the convergence of solutions as the parameter $\varepsilon \rightarrow 0$ (see Section 2). We generalize Fenichel's theorem for fast-reaction systems in infinite-dimensional Banach spaces. Under suitable assumptions we prove the existence of a C^1 -regular slow manifold that is close to the critical manifold in a suitable sense. Moreover, we show that the critical manifold attracts solutions and that the flows on the slow manifold converge to the flow on the critical manifold. To conclude this part we apply the theory to an example (see Section 3). A discussion on how to put the results into the larger context of invariant manifolds for PDEs is given at the end of the paper (see Section 4). Lastly we present a concise overview on the theory of C_0 semi-groups and their application to abstract Cauchy problems (see Appendix A). Moreover, we briefly present an alternative approach that uses a splitting type argument in both variables in order to control the ε^{-1} -terms (see Appendix B).

2. Fast reaction systems

In this section we analyse the general fast reaction system

$$\begin{aligned}\partial_t u^\varepsilon &= Au^\varepsilon + \frac{1}{\varepsilon}f(u^\varepsilon, v^\varepsilon), \\ \partial_t v^\varepsilon &= Bv^\varepsilon + g(u^\varepsilon, v^\varepsilon),\end{aligned}\tag{2.1}$$

for $\varepsilon > 0$ and show the convergence of the semi-flow of the solution of the ε -system to the semi-flow of the solution of the limit system as $\varepsilon \rightarrow 0$.

2.1. Function spaces and assumptions

We structure the assumptions into several parts.

Assumptions on the system:

- (S1) Let $0 < \varepsilon \ll 1$ be a parameter that encompasses the separation of time scales, where in the limit as $\varepsilon \rightarrow 0$ the PDE system (2.1) turns into an algebraic-differential equation.
- (S2) X, Y are Banach spaces and we denote by $X_1 = D(A)$ the domain of the operator A and $Y_1 = D(B)$ the domain of B respectively.
- (S3) We set the initial data $(u_0, v_0) \in X_1 \times Y_1$.

Assumptions on the operators A and B :

- (A1) A, B are closed linear operators, where $A : X \supset D(A) \rightarrow X$ generates the C_0 -semigroup $(e^{tA})_{t \geq 0} \subset \mathcal{B}(X)$ and $B : Y \supset D(B) \rightarrow Y$ generates the C_0 -semigroup $(e^{tB})_{t \geq 0} \subset \mathcal{B}(Y)$, respectively, where $\mathcal{B}(\cdot)$ denotes the space of bounded linear operators.
- (A2) The interpolation-extrapolation scales generated by (X, A) and (Y, B) are given by $(X_\alpha)_{\alpha \in [-1, \infty)}$ and $(Y_\alpha)_{\alpha \in [-1, \infty)}$ (for details regarding the definitions and basic properties of interpolation-extrapolation scales see Appendix A or [1]).
- (A3) Let $\gamma \in (0, 1]$ if $(e^{tA})_{t \geq 0}$ is analytic and $\gamma = 1$ otherwise. Similarly, we introduce $\delta \in (0, 1]$ if $(e^{tB})_{t \geq 0}$ is analytic and set $\delta = 1$ otherwise.
- (A4) There are constants $C_A, C_B, M_A, M_B > 0$, $\omega_A \in \mathbb{R}$ and $\omega_B \in \mathbb{R}$ such that

$$\begin{aligned} \|e^{tA}\|_{\mathcal{B}(X_1)} &\leq M_A e^{\omega_A t}, & \|e^{tA}\|_{\mathcal{B}(X_\gamma, X_1)} &\leq C_A t^{\gamma-1} e^{\omega_A t}, \\ \|e^{tB}\|_{\mathcal{B}(Y_1)} &\leq M_B e^{\omega_B t}, & \|e^{tB}\|_{\mathcal{B}(Y_\delta, Y_1)} &\leq C_B t^{\delta-1} e^{\omega_B t} \end{aligned}$$

hold for all $t \in (0, \infty)$.

Remark 2.1. We observe that we do not need the semigroups to be exponentially stable, i.e. we do not need to restrict ω_A, ω_B to be negative. An example of an analytic semigroup with positive growth bound is the shifted Laplacian $\Delta + \text{Id}$ acting on $H^2(\mathbb{T}^d)$. In this case we have $\omega = 1$.

Assumptions on the nonlinear functions f and g :

- (N1) Let $\alpha \in [0, 1]$ be such that $0 \leq \gamma - (1 - \alpha) \leq 1$. The nonlinearity $f : X_\alpha \times Y_\alpha \rightarrow X_{\gamma-(1-\alpha)}$ is a two-times continuously Fréchet differentiable function and $g : X_1 \times Y_1 \rightarrow Y_\delta$ is Fréchet differentiable. Moreover, there are constants $L_f, L_g > 0$ such that

$$\begin{aligned} \|Df(u, v)\|_{\mathcal{B}(X_1 \times Y_1, X_\gamma)} &\leq L_f, & \|Df(\tilde{u}, \tilde{v})\|_{\mathcal{B}(X_\alpha \times Y_\alpha, X_{\gamma-(1-\alpha)})} &\leq L_f, \\ \|Dg(u, v)\|_{\mathcal{B}(X_1 \times Y_1, Y_\delta)} &\leq L_g, \end{aligned} \tag{2.2}$$

for $(u, v) \in X_1 \times Y_1$ and $(\tilde{u}, \tilde{v}) \in X_\alpha \times Y_\alpha$.

- (N2) This implies that the following inequalities hold

$$\begin{aligned} \|f(u_1, v_1) - f(u_2, v_2)\|_{X_\gamma} &\leq L_f(\|u_1 - u_2\|_{X_1} + \|v_1 - v_2\|_{Y_1}), \\ \|f(\tilde{u}_1, \tilde{v}_1) - f(\tilde{u}_2, \tilde{v}_2)\|_{X_{\gamma-(1-\alpha)}} &\leq L_f(\|\tilde{u}_1 - \tilde{u}_2\|_{X_\alpha} + \|\tilde{v}_1 - \tilde{v}_2\|_{Y_\alpha}), \\ \|g(u_1, v_1) - g(u_2, v_2)\|_{Y_\delta} &\leq L_g(\|u_1 - u_2\|_{X_1} + \|v_1 - v_2\|_{Y_1}) \end{aligned}$$

for all $u_1, u_2 \in X_1$, $v_1, v_2 \in Y_1$. Here we assume that $f(u, v) \in X_\gamma$ if $(u, v) \in X_1 \times Y_1$.

(N3) For convenience, we assume that $f(0, 0) = 0$ and $g(0, 0) = 0$.

The next set of assumptions deals with the specific restrictions on the fast-reaction nonlinearity f .

Assumptions for the implicit function theorem to hold:

(IFT) Let $\mathcal{C} \subset X \times Y$ be a non-empty set such that for all $(u, v) \in \mathcal{C}$ we have $f(u, v) = 0$ and in addition the operator $D_u f(u, v) \equiv Df(\cdot, v)u : X \rightarrow X_\gamma$ is a Banach space isomorphism, i.e. a bounded invertible linear operator.

Lemma 2.2. *Let assumption (IFT) hold. Then, for each $(u_n, v_n) \in \mathcal{C}$ the implicit function theorem holds. That is, there exist open neighborhoods V_n of v_n and U_n of u_0 and a Fréchet differentiable function $h^{v_n} : V_n \rightarrow U_n$ such that $f(h^{v_n}(v_n), v_n) = 0$. Then h^{v_n} satisfies $h^{v_n}(v) = u$ and $f(h^{v_n}(v), v) = 0$ for all $(u, v) \in U_n \times V_n$. Moreover h^{v_n} has the same regularity as f .*

(N4) As we are interested in an attractive slow manifold we now only consider all the $(u, v) \in \mathcal{C}$ such that the operator $D_u f$ has its spectrum in the left half plane with an upper bound $\lambda_0 < 0$ on the real part and we denote this set by $\mathcal{C}_-$.

(N5) If there exist $(u_1, v_1), (u_2, v_2) \in \mathcal{C}_-$ such that the neighborhoods introduced in the implicit function theorem satisfy $U_1 \cap U_2 \neq \emptyset$ and $V_1 \cap V_2 \neq \emptyset$. Then, there exists $(u_3, v_3) \in \mathcal{C}_-$ such that $U_3 \subset U_1 \cap U_2$ and $V_3 \subset V_1 \cap V_2$. Moreover, we have that $h^{v_1}(v) = h^{v_2}(v)$ for all $v \in V_{y_1} \cap V_{y_2}$.

(N6) Next, we assume that $\mathcal{C}_-$ can be covered by a finite number of neighborhoods (U_n, V_n) and we write $h^n : V_n \rightarrow U_n$, where $h^n(v) = u$ for all $(u, v) \in U_n \times V_n$ for the representative function of the implicit function theorem of each connected neighborhood.

Remark 2.3. In many cases in the literature the fast-reaction term f satisfies the above assumptions with $\mathcal{C}_-$ being covered with one neighborhood. See Section 3.5 for an example of this behaviour. However, it can evidently happen, e.g., consider the nonlinearity $f(x, y) = 1 - x^2 - y^2$, where we need two charts to cover $\mathcal{C}_-$. As a third example we consider $f(x, y) = y \sin(x)$. Here, we need to restrict the variables to a compact set in order for the finite covering assumption to hold. Moreover, we observe that in this setting $\mathcal{C}_-$ consists of a finite number of disjoint subsets.

(N7) In order to keep the computations as clear as possible and to avoid dealing with chart transforms, we restrict the nonlinearity f to the case of one neighborhood (U_0, V_0) covering $\mathcal{C}_-$. And we denote the function from Lemma 2.2 by $h^0 : V_0 \rightarrow U_0$.

These are all the assumptions necessary for the existence of strict solutions and the convergence result of the following subsections. For the existence of the slow manifold result in Section 3 we require additional assumptions.

2.2. Existence of solutions

In this section we show the well-posedness of a general fast reaction system for $\varepsilon > 0$

$$\begin{aligned}\partial_t u^\varepsilon(t) &= Au^\varepsilon(t) + \frac{1}{\varepsilon} f(u^\varepsilon(t), v^\varepsilon(t)), \\ \partial_t v^\varepsilon(t) &= Bv^\varepsilon(t) + g(u^\varepsilon(t), v^\varepsilon(t)), \\ u^\varepsilon(0) &= u_0, \quad v^\varepsilon(0) = v_0,\end{aligned}\tag{2.3}$$

and its limit system when $\varepsilon = 0$

$$\begin{aligned}0 &= f(u^0(t), v^0(t)), \\ \partial_t v^0(t) &= Bv^0(t) + g(u^0(t), v^0(t)), \\ u^0(0) &= u_0, \quad v^0(0) = v_0.\end{aligned}\tag{2.4}$$

This system is the standard form for fast reaction systems found in the literature. And in the subsequent sections all results will be posed for systems (2.3) and (2.4). However, the system is in non-standard form when it comes to the fast-slow analysis. To this end we introduce a modified fast reaction system that allows for a better control on the ε -dependent terms in the proofs.

$$\begin{aligned}\partial_t u^\varepsilon &= \frac{1}{\varepsilon} \tilde{A}_\varepsilon u^\varepsilon + \frac{1}{\varepsilon} \tilde{f}(u^\varepsilon, v^\varepsilon), \\ \partial_t v^\varepsilon &= Bv^\varepsilon + g(u^\varepsilon, v^\varepsilon), \\ u^\varepsilon(0) &= u_0, \quad v^\varepsilon(0) = v_0,\end{aligned}\tag{2.5}$$

where $\tilde{A}_\varepsilon := \varepsilon A + D_u f(u^0, v^0)$ and $\tilde{f}(u, v) := f(u, v) - D_u f(u^0, v^0)u$.

Remark 2.4. For simplicity we assume that for systems (2.3)-(2.5) we have the same initial data $(u_0, v_0) \in X_1 \times Y_1 \cap S_0$. However, it is also possible to consider different sets of initial data for the two systems. In this case we would require that $(u_0^\varepsilon, v_0^\varepsilon) \rightarrow (u_0^0, v_0^0)$ in $X_1 \times Y_1$ as $\varepsilon \rightarrow 0$.

Lemma 2.5. *The function $\tilde{f}$ has the same regularity as f and its Lipschitz constant satisfies*

$$L_{\tilde{f}} \leq L_f + \sup_{0 \leq t < \infty} \|D_u f(u^0(t), v^0(t))\|_{\mathcal{B}(X_1 \times Y_1, X_\gamma)}.$$

Now we can state the existence result for system (2.4).

Proposition 2.6. *Let $\varepsilon = 0$ and let Assumptions (S1)-(N7) hold. In addition, we assume that*

$$(N8) \quad L_{\tilde{f}} \inf_{0 \leq t < \infty} \|(D_u f(u^0(t), v^0(t)))^{-1}\|_{\mathcal{B}(X_1 \times Y_1, X_\gamma)} < 1 \quad \text{holds.}$$

Then, there exists a unique strict solution $(u^0, v^0) \in C^1([0, \infty); X \times Y) \cap C([0, \infty); X_1 \times Y_1)$ to equation (2.4).

Proof. The first part of the proof is to apply the implicit function theorem. From the assumptions on the initial data and the nonlinearity f we have that $f(u_0, v_0) = 0$. Then, following the observations from the previous section we can apply the implicit function theorem at $(u_0, v_0) \in X_1 \times Y_1$ and

obtain the existence of a neighborhood $(U_0, V_0) \subset \mathcal{C}_-$ and a Fréchet differentiable function $h^0 : V_0 \subset Y_1 \rightarrow U_0 \subset X_1$ such that $h^0(y) = x$ for all $(x, y) \in U_0 \times V_0$. Moreover, as $D_u f(u_0, v_0)h^0(y) + \tilde{f}(h^0(y), y) = 0$ we can write $h^0(y) = -D_u f(u_0, v_0)^{-1} \tilde{f}(h^0(y), y)$. Next, we prove that h^0 is indeed Lipschitz continuous. To see this, let $y_1, y_2 \in V_0$ and we compute

$$\begin{aligned} \|h^0(y_1) - h^0(y_2)\|_{X_1} &\leq \\ &\leq \|D_u f(u_0, v_0)^{-1}(\tilde{f}(h^0(y_1), y_1) - \tilde{f}(h^0(y_2), y_2))\|_{X_1} \\ &\leq L_{\tilde{f}}\|(D_u f(u_0, v_0))^{-1}\|_{\mathcal{B}(X_\gamma, X_1)}(\|h^0(y_1) - h^0(y_2)\|_{X_1} + \|y_1 - y_2\|_{Y_1}) \end{aligned}$$

which yields

$$\begin{aligned} \|h^0(y_1) - h^0(y_2)\|_{X_1} &\leq L_{\tilde{f}}\|(D_u f(u_0, v_0))^{-1}\|_{\mathcal{B}(X_\gamma, X_1)} \\ &\quad \times (1 - L_{\tilde{f}}\|(D_u f(u_0, v_0))^{-1}\|_{\mathcal{B}(X_\gamma, X_1)})^{-1}\|y_1 - y_2\|_{Y_1} \end{aligned}$$

and thus

$$\|h^0(y_1) - h^0(y_2)\|_{X_1} \leq L_{h^0}\|y_1 - y_2\|_{Y_1} \quad (2.6)$$

Thus, the mapping $g \circ h^0 : Y_1 \rightarrow Y_\delta$, $y \mapsto g(h^0(y), y)$ is Lipschitz continuous and satisfies

$$\begin{aligned} \|g(h^0(y_1), y_1) - g(h^0(y_2), y_2)\|_{Y_\delta} &\leq L_g\|h^0(y_1) - h^0(y_2)\|_{Y_1} + L_g\|y_1 - y_2\|_{Y_1} \\ &\leq L_g(1 + L_{h^0})\|y_1 - y_2\|_{Y_1}. \end{aligned}$$

Then, there exists a unique strict solution $v^0 \in C^1([0, \infty); Y) \cap C([0, \infty); Y_1)$ to

$$\partial_t v^0 = Bv^0 + g(h^0(v^0), v^0), \quad v^0(0) = v_0.$$

This follows by using the variation of constants formula and a suitable weighted space, similar to the one in [45], by proving that the operator

$$\mathcal{L}_B(v^0) := e^{Bt}v_0 + \int_0^t e^{B(t-s)}g(h^0(v^0), v^0)ds$$

has a fixed point. Now, setting $u^0(t) := h^0(v^0(t))$ we obtain that

$$u^0 \in C^1([0, \infty); X) \cap C([0, \infty); X_1).$$

Thus, $(u^0, v^0) \in C^1([0, \infty); X \times Y) \cap C([0, \infty); X_1 \times Y_1)$ is the unique strict solution of the limit system (2.4) with $\varepsilon = 0$. $\square$

Remark 2.7. The mapping

$$h^0 : V_0 \subset Y_1 \rightarrow X_1, \quad y \mapsto h^0(y),$$

satisfying $f(h^0(y), y) = 0$, introduced in the previous proof describes the critical manifold S_0 over Y_1 given by

$$S_0 := \{(h^0(y), y) : y \in Y_1\} \subset X \times Y.$$

Proposition 2.8. *Let $\varepsilon > 0$ be small. Then, there exists a unique strict solution $(u^\varepsilon, v^\varepsilon)$ of the system (2.3), i.e the solution satisfies $(u^\varepsilon, v^\varepsilon) \in C^1([0, \infty); X \times Y) \cap C([0, \infty); X_1 \times Y_1)$.*

Proof. The idea of the proof is to construct a contracting solution operator and apply Banach's fixed point theorem on a suitable space. The space we consider is $C_b([0, \infty), e^{\eta t}; X_1 \times Y_1)$, i.e. the space of all $(u, v) \in C([0, \infty); X_1 \times Y_1)$ such that

$$\|(u, v)\|_{C_b([0, \infty), e^{\eta t}; X_1 \times Y_1)} := \sup_{t \geq 0} e^{-\eta t} (\|u(t)\|_{X_1} + \|v(t)\|_{Y_1}) < \infty,$$

where $\eta \geq 0$ and the precise value will be determined later. On this space we now aim to show that the operator

$$\mathcal{L}(u, v) := \begin{pmatrix} e^{tA}u_0 & + \varepsilon^{-1} \int_0^t e^{(t-s)A} f(u(s), v(s)) \, ds \\ e^{tB}v_0 & + \int_0^t e^{(t-s)B} g(u(s), v(s)) \, ds. \end{pmatrix}$$

is a contraction. Thus, estimating $\|\mathcal{L}(u, v) - \mathcal{L}(\tilde{u}, \tilde{v})\|_{X_1 \times Y_1}$ yields

$$\begin{aligned} \sup_{t \geq 0} e^{-\eta t} \left\| \varepsilon^{-1} \int_0^t e^{(t-s)A} [f(u(s), v(s)) - f(\tilde{u}(s), \tilde{v}(s))] \, ds \right\|_{X_1} &\leq \\ &\leq L_f C_A \varepsilon^{-1} \int_0^t \frac{e^{(t-s)(\omega_A - \eta)}}{(t-s)^{1-\gamma}} \, ds \| (u(s), v(s)) - (\tilde{u}(s), \tilde{v}(s)) \|_{C_b([0, \infty), e^{\eta t}; X_1 \times Y_1)} \\ &\leq L_f C_A \varepsilon^{-1} \frac{\Gamma(\gamma)}{(\eta - \omega_A)^\gamma} \| (u(s), v(s)) - (\tilde{u}(s), \tilde{v}(s)) \|_{C_b([0, \infty), e^{\eta t}; X_1 \times Y_1)} \end{aligned}$$

and

$$\begin{aligned} \sup_{t \geq 0} e^{-\eta t} \left\| \int_0^t e^{(t-s)B} [g(u(s), v(s)) - g(\tilde{u}(s), \tilde{v}(s))] \, ds \right\|_{Y_1} &\leq \\ &\leq L_g C_B \int_0^t \frac{e^{(t-s)(\omega_B - \eta)}}{(t-s)^{1-\delta}} \, ds \| (u(s), v(s)) - (\tilde{u}(s), \tilde{v}(s)) \|_{C_b([0, \infty), e^{\eta t}; X_1 \times Y_1)} \\ &\leq L_g C_B \frac{\Gamma(\delta)}{(\eta - \omega_B)^\delta} \| (u(s), v(s)) - (\tilde{u}(s), \tilde{v}(s)) \|_{C_b([0, \infty), e^{\eta t}; X_1 \times Y_1)}. \end{aligned}$$

For η large enough we have that $\mathcal{L}$ is a contraction and thus there exists a unique fixed point

$$(u_*^\varepsilon, v_*^\varepsilon) \in C_b([0, \infty), e^{\eta t}; X_1 \times Y_1).$$

Let $(u^\varepsilon, v^\varepsilon)$ be this fixed point. Then, we have that

$$\begin{aligned} u^\varepsilon(t) &= e^{tA}u_0 + \frac{1}{\varepsilon} \int_0^t e^{(t-s)A} f(u^\varepsilon(s), v^\varepsilon(s)) \, ds, \\ v^\varepsilon(t) &= e^{tB}v_0 + \int_0^t e^{(t-s)B} g(u^\varepsilon(s), v^\varepsilon(s)) \, ds \end{aligned}$$

which implies that

$$\begin{aligned} u^\varepsilon(t) &= u_0 + A \int_0^t u^\varepsilon(s) \, ds + \frac{1}{\varepsilon} \int_0^t f(u^\varepsilon(s), v^\varepsilon(s)) \, ds, \\ v^\varepsilon(t) &= v_0 + B \int_0^t v^\varepsilon(s) \, ds + \int_0^t g(u^\varepsilon(s), v^\varepsilon(s)) \, ds \end{aligned}$$

see [45] for details. Thus, for all $t \in [0, \infty)$ we conclude that

$$\begin{aligned}\partial_t u^\varepsilon(t) &= \lim_{h \rightarrow 0} \frac{u^\varepsilon(t+h) - u^\varepsilon(t)}{h} \\ &= \lim_{h \rightarrow 0} \left[\int_t^{t+h} A u^\varepsilon(s) \, ds + \frac{1}{\varepsilon} \int_t^{t+h} f(u^\varepsilon(s), v^\varepsilon(s)) \, ds \right]\end{aligned}$$

is well-defined and in the limit

$$\partial_t u^\varepsilon(t) = A u^\varepsilon(t) + \frac{1}{\varepsilon} f(u^\varepsilon(t), v^\varepsilon(t)),$$

where the convergence holds in X as $A u^\varepsilon, f(u^\varepsilon, v^\varepsilon) \in C([0, \infty); X)$. With the same idea we can show that $\partial_t v^\varepsilon \in C([0, \infty); Y)$. Hence, we have shown that

$$(u^\varepsilon, v^\varepsilon) \in C([0, \infty); X_1 \times Y_1) \cap C^1([0, \infty); X \times Y)$$

is the unique solution to (2.3) for $\varepsilon > 0$. $\square$

We conclude this section with the following result.

Lemma 2.9. *Let Assumptions (S1)-(N8) hold. Then, applying the perturbation result for semigroups (see Appendix A) yields that the operator $\tilde{A}_\varepsilon$ satisfies the following estimates*

$$\|e^{t\tilde{A}_\varepsilon}\|_{\mathcal{B}(X_1)} \leq M_{\tilde{A}} e^{(\varepsilon\omega_A + \lambda_0)t}, \quad \|e^{t\tilde{A}_\varepsilon}\|_{\mathcal{B}(X_\gamma, X_1)} \leq C_{\tilde{A}} t^{\gamma-1} e^{(\varepsilon\omega_A + \lambda_0)t}$$

for some constants $M_{\tilde{A}}, C_{\tilde{A}} > 0$ and all $t > 0$.

This implies that Proposition 2.8 also holds for the modified fast reaction system (2.5).

2.3. Approximation by slow flow

The main result of this subsection is to show the convergence of the solutions of the full system (2.3) to the solutions of the limit system (2.4) as $\varepsilon \rightarrow 0$ via the perturbed system (2.5). A key step in the proof will be to estimate the convergence in the fast component. To this end we introduce two approximate systems in u , namely $\tilde{u}^\varepsilon$ and $u^{\varepsilon,0}$, where $\tilde{u}^\varepsilon$ is the solution of

$$\partial_t \tilde{u}^\varepsilon = \frac{1}{\varepsilon} \tilde{A}_\varepsilon u^\varepsilon + \frac{1}{\varepsilon} \tilde{f}(\tilde{u}^\varepsilon, v^0) \quad (2.7)$$

and $u^{\varepsilon,0}$ is the solution of

$$\partial_t u^{\varepsilon,0} = \frac{1}{\varepsilon} \tilde{A}_\varepsilon u^{\varepsilon,0} + \frac{1}{\varepsilon} \tilde{f}(u^{\varepsilon,0}, v^0) + \partial_t (\tilde{A}_\varepsilon)^{-1} \tilde{f}(h^0(v^0), v^0) \quad (2.8)$$

and where in both cases v^0 satisfies the second equation of the system (2.4). The expression $(\tilde{A}_\varepsilon)^{-1}$ is well defined since the resolvent of A is positive and the spectrum of $D_u f(h^0(v^0), v^0)$ is by assumption completely on the left half plane. The existence of solutions follows from adapting the proofs of Propositions 2.6 and 2.8.

We start by proving an estimate on the perturbed operator $\tilde{A}_\varepsilon$, that explains the loss of regularity that will be encountered in the proceeding estimates and is the main novelty when comparing it to the results in [35].

Lemma 2.10. *Let the operators A and $\tilde{A}_\varepsilon$ satisfy Assumption (A1)-(A3) and Lemma 2.9. Then the following holds*

$$\|A(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X_\beta, X_\alpha)} \leq \varepsilon^{-(1+\beta-\alpha)} C. \quad (2.9)$$

Proof. Let λ_n be the eigenvalue corresponding to the eigenvector e_n of the operator A , such that the eigenvectors form an orthonormal basis of the Banach space X . In addition, we assume that $\varepsilon\lambda_n + \lambda(t)$, where $\sup_t \lambda(t) < \lambda_0 < 0$ is the eigenvalue of $\tilde{A}_\varepsilon$ corresponding to the eigenvector e_n . Setting $\mu = \alpha - \beta$ and denoting by $\text{pr}_n x$ the projection of $x \in X$ onto the eigenvector e_n we obtain

$$\begin{aligned} & \|A(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X_\beta, X_\alpha)}^2 = \\ &= \sup_{\|x\|_{X_\alpha}=1} \|A(\tilde{A}_\varepsilon)^{-1}x\|_{X_\beta}^2 \leq \sup_{\|x\|_{X_\alpha}=1} \sum_{n \in \mathbb{N}} \frac{\lambda_n^2}{(-\varepsilon\lambda_n + \lambda(t))^2} \lambda_n^{2\beta} \|\text{pr}_n x\|_X^2 \\ &\leq \sup_{\|x\|_{X_\alpha}=1} \sum_{n \in \mathbb{N}} \frac{\lambda_n^2}{(-\varepsilon\lambda_n + \lambda_0)^2} \lambda_n^{2\beta} \|\text{pr}_n x\|_X^2 \\ &\leq \sup_{\|x\|_{X_\alpha}=1} \left(\sum_{n: |\lambda_n| \leq \varepsilon^{-1}} \frac{\lambda_n^{2-2\mu} \lambda_n^{2\alpha}}{(-\varepsilon\lambda_n + \lambda_0)^2} \|\text{pr}_n x\|_X^2 + \right. \\ &\quad \left. + \sum_{n: |\lambda_n| > \varepsilon^{-1}} \frac{\lambda_n^{2-2\mu} \lambda_n^{2\alpha}}{(-\varepsilon\lambda_n + \lambda_0)^2} \|\text{pr}_n x\|_X^2 \right) \\ &\leq \sup_{\|x\|_{X_\alpha}=1} \left(\sum_{n: |\lambda_n| \leq \varepsilon^{-1}} \frac{\lambda_n^{2\alpha}}{(-\varepsilon\lambda_n^\mu + \lambda_0 \lambda^{2\mu-2})^2} \|\text{pr}_n x\|_X^2 + \right. \\ &\quad \left. + \sum_{n: |\lambda_n| > \varepsilon^{-1}} \frac{\lambda_n^{2-2\mu} \lambda_n^{2\alpha}}{(-\varepsilon\lambda_n + \lambda_0)^2} \|\text{pr}_n x\|_X^2 \right) \\ &\leq \sup_{\|x\|_{X_\alpha}=1} \left(\sum_{n: |\lambda_n| \leq \varepsilon^{-1}} \frac{\lambda_n^{2\alpha}}{(\lambda_0 \lambda^{2\mu-2})^2} \|\text{pr}_n x\|_X^2 + \sum_{n: |\lambda_n| > \varepsilon^{-1}} \frac{\varepsilon^{2\mu} \lambda_n^{2\alpha}}{\varepsilon^2} \|\text{pr}_n x\|_X^2 \right) \\ &\leq C + \varepsilon^{-2(1-\mu)} \leq \varepsilon^{-2(1-\mu)} C. \end{aligned}$$

□

Thus, only for $\alpha - \beta < 1$ the norm of $\varepsilon\|A(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X_\beta, X_\alpha)}$ can be controlled by ε^μ , with $\mu > 0$. Therefore, we require the estimates are posed in X_α with $0 \leq \alpha < 1$, whereas the initial data is in X_1 . This marks also a further difference to the results in [35].

Now, we present three auxiliary results that connect the fast components of systems (2.3) and (2.4) with the approximate equations (2.7) and (2.8).

Proposition 2.11. *Let u^ε be the solution of (2.3) and $\tilde{u}^\varepsilon$ be the solution of (2.7). Then we have the estimate*

$$\|u^\varepsilon(t) - \tilde{u}^\varepsilon(t)\|_{X_\alpha} \leq C_A L_{\tilde{f}} C_1 \frac{\Gamma(\gamma)}{|\varepsilon\omega_A - \lambda_0|^\gamma} \sup_{0 \leq s \leq t} \|v^\varepsilon(s) - v^0(s)\|_{Y_1}, \quad (2.10)$$

where $C_1 = e^{C_A L_{\tilde{f}} \frac{\Gamma(\gamma)}{|\varepsilon\omega_A - \lambda_0|^\gamma}}$.

Proof. Using variation of constants yields

$$\begin{aligned} \|u^\varepsilon(t) - \tilde{u}^\varepsilon(t)\|_{X_\alpha} &\leq \left\| \varepsilon^{-1} \int_0^t e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon} (\tilde{f}(u^\varepsilon, v^\varepsilon) - \tilde{f}(\tilde{u}^\varepsilon, v^0)) \, ds \right\|_{X_\alpha} \\ &\leq L_{\tilde{f}} C_A \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} (\|u^\varepsilon(s) - \tilde{u}^\varepsilon(s)\|_{X_\alpha} + \|v^\varepsilon(s) - v^0(s)\|_{Y_1}) \, ds \\ &\leq \frac{L_{\tilde{f}} C_A \Gamma(\gamma)}{|\varepsilon\omega_A - \lambda_0|^\gamma} \sup_{0 \leq s \leq t} \|v^\varepsilon(s) - v^0(s)\|_{Y_1} \\ &\quad + L_{\tilde{f}} C_A \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} \|u^\varepsilon(s) - \tilde{u}^\varepsilon(s)\|_{X_\alpha} \, ds. \end{aligned}$$

Then, by applying Gronwall's inequality the result follows

$$\|u^\varepsilon(t) - \tilde{u}^\varepsilon(t)\|_{X_\alpha} \leq C_A L_{\tilde{f}} C_1 \frac{\Gamma(\gamma)}{|\varepsilon\omega_A - \lambda_0|^\gamma} \sup_{0 \leq s \leq t} \|v^\varepsilon(s) - v^0(s)\|_{Y_1},$$

where $C_1 = e^{C_A L_{\tilde{f}} \frac{\Gamma(\gamma)}{|\varepsilon\omega_A - \lambda_0|^\gamma}}$.

□

Proposition 2.12. *Let $\tilde{u}^\varepsilon$ be the solution of (2.7) and $u^{\varepsilon,0}$ be the solution of (2.8). Then we have the estimate*

$$\|\tilde{u}^\varepsilon(t) - u^{\varepsilon,0}(t)\|_{X_\alpha} \leq \varepsilon \frac{C C_A L_{\tilde{f}} \Gamma(\gamma)}{|\varepsilon\omega_A + \lambda_0|^\gamma} (\varepsilon^{-\gamma+(1-\alpha)} + e^{\omega_B t} \|v_0\|_{Y_1}) e^{\omega_B t} \|v_0\|_{Y_1}. \quad (2.11)$$

Proof. Again using variation of constants yields

$$\begin{aligned} \|\tilde{u}^\varepsilon(t) - u^{\varepsilon,0}(t)\|_{X_\alpha} &\leq \left\| \varepsilon^{-1} \int_0^t e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon} (\tilde{f}(\tilde{u}^\varepsilon, v^0) - \tilde{f}(u^{\varepsilon,0}, v^0) - \varepsilon \partial_s (\tilde{A}_\varepsilon)^{-1} \tilde{f}(h^0(v^0), v^0)) \, ds \right\|_{X_\alpha}. \end{aligned}$$

Then, we can estimate the first term with the standard Lipschitz estimate and the second term is split into two parts. This then yields

$$\begin{aligned} &\leq C_A L_{\tilde{f}} \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A - \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} \|\tilde{u}^\varepsilon(s) - u^{\varepsilon,0}(s)\|_{X_\alpha} \, ds \\ &\quad + \varepsilon C_A \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A - \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} \|(\tilde{A}_\varepsilon)^{-1} \partial_s \tilde{f}(h^0(v^0), v^0)\|_{X_{\gamma-(1-\alpha)}} \, ds \\ &\quad + \varepsilon C_A L_{\tilde{f}} \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A - \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} \|\partial_s (\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X_{\gamma-(1-\alpha)})} (1 + L_{h^0}) \|v^0\|_{Y_1} \, ds. \end{aligned}$$

The key challenge is now to bound each term in the appropriate spaces.

$$\leq C_A L_{\tilde{f}} \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A - \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} \|\tilde{u}^\varepsilon(s) - u^{\varepsilon,0}(s)\|_{X_\alpha} \, ds$$

$$\begin{aligned}
& + \frac{\varepsilon C_A L_{\tilde{f}}}{|\varepsilon \omega_A + \lambda_0|^\gamma} \|(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X, X_{\gamma-(1-\alpha)})} \|D\tilde{f}(h^0(v^0), v^0)\|_{\mathcal{B}(X \times Y, X_{\gamma-(1-\alpha)})} \\
& \quad \times (\|\partial_t h^0(v^0)\|_{L^\infty([0,t], X)} + \|\partial_t v^0\|_{L^\infty([0,\infty), Y)}) \\
& + \frac{\varepsilon C_A L_{\tilde{f}}(1 + L_{h^0})}{|\varepsilon \omega_A + \lambda_0|^\gamma} \|(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X, X_{\gamma-(1-\alpha)})} \|(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X)} \\
& \quad \times \|D^2 f(h^0(v^0), v^0)\|_{\mathcal{B}(X \times Y, X)} \\
& \quad \times (\|\partial_t h^0(v^0)\|_{L^\infty([0,t], X)} + \|\partial_t v^0\|_{L^\infty([0,t], Y)}) \|v^0\|_{Y_1} \\
& \leq C_A L_{\tilde{f}} \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon \omega_A - \lambda_0)}}{\varepsilon^\gamma (t-s)^{1-\gamma}} \|\tilde{u}^\varepsilon(s) - u^{\varepsilon,0}(s)\|_{X_\alpha} \, ds \\
& + \frac{\varepsilon C C_A L_{\tilde{f}}}{|\varepsilon \omega_A + \lambda_0|^\gamma} (\varepsilon^{-\gamma+(1-\alpha)} + e^{\omega_B t} \|v_0\|_{Y_1}) \\
& \quad \times (\|\partial_t h^0(v^0)\|_{L^\infty([0,t], X)} + \|\partial_t v^0\|_{L^\infty([0,t], Y)})
\end{aligned}$$

where we used the estimate $\|(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X, X_{\gamma-(1-\alpha)})} \leq \varepsilon^{-\gamma+(1-\alpha)}$, see Lemma 2.10 for more details. Then, by applying Gronwall's inequality we obtain

$$\begin{aligned}
\|\tilde{u}^\varepsilon(t) - u^{\varepsilon,0}(t)\|_{X_\alpha} & \leq \varepsilon \frac{C C_A L_{\tilde{f}} \Gamma(\gamma)}{|\varepsilon \omega_A + \lambda_0|^\gamma} (\varepsilon^{-\gamma+(1-\alpha)} + e^{\omega_B t} \|v_0\|_{Y_1}) \\
& \quad \times (\|\partial_t h^0(v^0)\|_{L^\infty([0,t], X)} + \|\partial_t v^0\|_{L^\infty([0,t], Y)}).
\end{aligned}$$

To improve this estimate we note that

$$\|\partial_t h^0(v^0)\|_{L^\infty([0,t], X)} \leq \|h^0(v^0)\|_{C_b^1([0,t], X)} \leq L_{h^0} \|v^0\|_{C_b^1([0,t], Y)}$$

and by applying inequalities (2.14) and (2.15) we obtain

$$\begin{aligned}
\|\partial_t h^0(v^0)\|_{L^\infty([0,t], X)} & \leq C e^{\omega_B t} (\|v_0\|_{Y_1} + \|B v_0\|_Y + L_g(L_{h^0} + 1) \|v_0\|_Y) \\
& \leq C e^{\omega_B t} \|v_0\|_{Y_1}.
\end{aligned}$$

Hence, we can conclude that

$$\|\tilde{u}^\varepsilon(t) - u^{\varepsilon,0}(t)\|_{X_\alpha} \leq \varepsilon \frac{C C_A L_{\tilde{f}} \Gamma(\gamma)}{|\varepsilon \omega_A + \lambda_0|^\gamma} (\varepsilon^{-\gamma+(1-\alpha)} + e^{\omega_B t} \|v_0\|_{Y_1}) e^{\omega_B t} \|v_0\|_{Y_1}.$$

□

Proposition 2.13. *Let $u^{\varepsilon,0}$ be the solution of (2.8) and let $u^0 = h^0(v^0)$ be the solution of (2.4). Then we have the estimate*

$$\begin{aligned}
& \|u^{\varepsilon,0}(t) - h^0(v^0(t))\|_{X_\alpha} \leq \\
& \leq M_A \varepsilon^{-\gamma} e^{\varepsilon^{-1}(\varepsilon \omega_A + \lambda_0)t} t^\gamma \|u_0 - h^0(v_0)\|_{X_\alpha} + \varepsilon^{1-\alpha} C e^{\omega_B t} \|v_0\|_{Y_1} \\
& + C \varepsilon^{-\gamma+(1-\alpha)} e^{\varepsilon^{-1}(\varepsilon \omega_A + \lambda_0)t} t^\gamma \|v_0\|_{Y_1}.
\end{aligned} \tag{2.12}$$

Proof. Using the variation of constants formula we have

$$\begin{aligned}
& \|u^{\varepsilon,0}(t) - h^0(v^0(t))\|_{X_\alpha} = \\
& = \left\| e^{t\varepsilon^{-1}\tilde{A}_\varepsilon} u_0 + \varepsilon^{-1} \int_0^t e^{(t-s)\varepsilon^{-1}\tilde{A}_\varepsilon} \tilde{f}(u^{\varepsilon,0}(s), v^0(s)) \, ds \right.
\end{aligned}$$

$$+ \int_0^t e^{(t-s)\varepsilon^{-1}\tilde{A}_\varepsilon} \partial_s (\tilde{A}_\varepsilon)^{-1} \tilde{f}(h^0(v^0(s)), v^0(s)) \, ds - h^0(v^0(t)) \Big\|_{X_\alpha}.$$

Then, integration by parts yields

$$= \left\| e^{t\varepsilon^{-1}\tilde{A}_\varepsilon} u_0 + \varepsilon^{-1} \int_0^t e^{(t-s)\varepsilon^{-1}\tilde{A}_\varepsilon} (\tilde{f}(u^{\varepsilon,0}(s), v^0(s)) - \tilde{f}(h^0(v^0(s)), v^0(s))) \, ds \right. \\ \left. + (\tilde{A}_\varepsilon)^{-1} \tilde{f}(h^0(v^0(t)), v^0(t)) - h^0(v^0(t)) - e^{t\varepsilon^{-1}\tilde{A}_\varepsilon} (\tilde{A}_\varepsilon)^{-1} \tilde{f}(h^0(v_0), v_0) \right\|_{X_\alpha}.$$

Recalling that $0 = D_x f(h^0(v^0), v^0) h^0(v^0) + \tilde{f}(h^0(v^0), v^0)$ we estimate

$$\|u^{\varepsilon,0}(t) - h^0(v^0(t))\|_{X_\alpha} \leq \\ \leq C_A L_{\tilde{f}} \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma (t-s)^{1-\gamma}} \| (u^{\varepsilon,0}(s) - h^0(v^0(s))) \|_{X_\alpha} \, ds \\ + M_A e^{\varepsilon^{-1}(\varepsilon\omega_A - \lambda_0)t} \|u_0 - h^0(v_0)\|_{X_\alpha} \\ + M_A e^{\varepsilon^{-1}(\varepsilon\omega_A - \lambda_0)t} \|h^0(v_0) - (\tilde{A}_\varepsilon)^{-1} D_x f(h^0(v^0), v^0) h^0(v_0)\|_{X_\alpha} \\ + \|h^0(v^0(t)) - (\tilde{A}_\varepsilon)^{-1} D_x f(h^0(v^0), v^0) h^0(v^0(t))\|_{X_\alpha}.$$

Here, we note the additional terms in the estimate when compared with the result in [35, Thm. 4.13]. This is due to the fact that $D_x f(h^0(v^0), v^0) (\tilde{A}_\varepsilon)^{-1}$ does not equal the identity operator Id but a perturbation of the latter. To be more precise, let $z \in X_1$, then we have that

$$\|(\text{Id} - (\tilde{A}_\varepsilon)^{-1} D_x f(h^0(v^0), v^0)) z\|_{X_\alpha} = \varepsilon \|(\tilde{A}_\varepsilon)^{-1} A z\|_{X_\alpha}$$

and we can write

$$(\tilde{A}_\varepsilon)^{-1} D_x f(h^0(v^0), v^0) = \text{Id} - \varepsilon J,$$

where the operator $J := (\tilde{A}_\varepsilon)^{-1} A : X_1 \rightarrow X_\alpha$ is bounded by $\|J\|_{\mathcal{B}(X_\alpha, X_1)} \leq C\varepsilon^{-\alpha}$, see Lemma 2.10. Thus,

$$\|u^{\varepsilon,0}(t) - h^0(v^0(t))\|_{X_\alpha} \leq \\ \leq C_A L_{\tilde{f}} \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma (t-s)^{1-\gamma}} \| (u^{\varepsilon,0}(s) - h^0(v^0(s))) \|_{X_\alpha} \, ds \\ + M_A e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t} \|u_0 - h^0(v_0)\|_{X_\alpha} \\ + \varepsilon M_A L_{h^0} e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t} \|J\|_{\mathcal{B}(X_\alpha, X_1)} \|v_0\|_{Y_1} \\ + \varepsilon L_{h^0} \|J\|_{\mathcal{B}(X_\alpha, X_1)} \|v^0(t)\|_{Y_1}.$$

Using estimate (2.14) and applying Gronwall's inequality yields

$$\|u^{\varepsilon,0}(t) - h^0(v^0(t))\|_{X_\alpha} \leq \\ \leq M_A e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t} (\|u_0 - h^0(v_0)\|_{X_\alpha} + \varepsilon^{1-\alpha} C \|v_0\|_{Y_1}) + \varepsilon^{1-\alpha} C e^{\omega_B t} \|v_0\|_{Y_1} \\ + C_A L_{\tilde{f}} M_A \|u_0 - h^0(v_0)\|_{X_\alpha} e^{\frac{\Gamma(\gamma)}{|\varepsilon\omega_A + \lambda_0|^\gamma}} \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma (t-s)^{1-\gamma}} e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)s} \, ds$$

$$\begin{aligned}
& + \varepsilon^{1-\alpha} C C_A L_{\tilde{f}} M_A L_{h^0} \|v_0\|_{Y_1} e^{\frac{\Gamma(\gamma)}{|\varepsilon\omega_A + \lambda_0|^\gamma}} \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma (t-s)^{1-\gamma}} e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)s} ds \\
& + \varepsilon^{1-\alpha} C_A L_{\tilde{f}} C L_{h^0} \|v_0\|_{Y_1} e^{\frac{\Gamma(\gamma)}{|\varepsilon\omega_A + \lambda_0|^\gamma}} \int_0^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma (t-s)^{1-\gamma}} e^{\omega_B s} ds.
\end{aligned}$$

This inequality can be further simplified as follows

$$\begin{aligned}
& \|u^{\varepsilon,0}(t) - h^0(v^0(t))\|_{X_\alpha} \leq \\
& \leq M_A \varepsilon^{-\gamma} e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t} t^\gamma \|u_0 - h^0(v_0)\|_{X_\alpha} + \varepsilon^{1-\alpha} C e^{\omega_B t} \|v_0\|_{Y_1} \\
& + C \varepsilon^{-\gamma+(1-\alpha)} e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t} t^\gamma \|v_0\|_{Y_1},
\end{aligned} \tag{2.13}$$

where we observe that $\lim_{\varepsilon \rightarrow 0} \varepsilon^{-\gamma} e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t} = 0$ for $t > 0$. $\square$

Theorem 2.14. *There are constants $C_1, C_2 > 0$ such that the solution (u^0, v^0) of system (2.4) and $(u^\varepsilon, v^\varepsilon)$ of system (2.3) satisfy*

$$\begin{aligned}
\left\| \begin{pmatrix} u^\varepsilon(t) - u^0(t) \\ v^\varepsilon(t) - v^0(t) \end{pmatrix} \right\|_{X_\alpha \times Y_1} & \leq C_1 e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t} \|u_0 - h^0(v_0)\|_{X_\alpha} \\
& + C_2 \varepsilon^{1-\alpha} (e^{\omega_B t} + \varepsilon^{-\gamma} t^\gamma e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t}) \|v_0\|_{Y_1}
\end{aligned}$$

for all $t \in [0, \infty)$ and all $\varepsilon > 0$, where $0 \leq \alpha < 1$.

Proof. To control the ε -dependent terms we will work with the modified system (2.5). Then, using the variation of constants formula the solutions to systems (2.5) and (2.4) can be written as

$$\begin{aligned}
u^\varepsilon(t) &= e^{\varepsilon^{-1}t\tilde{A}_\varepsilon} u_0 + \varepsilon^{-1} \int_0^t e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon} \tilde{f}(u^\varepsilon(s), v^\varepsilon(s)) ds, \\
v^\varepsilon(t) &= e^{tB} v_0 + \int_0^t e^{(t-s)B} g(u^\varepsilon(s), v^\varepsilon(s)) ds,
\end{aligned}$$

and

$$\begin{aligned}
0 &= D_u f(h^0(v^0(t)), v^0(t)) h^0(v^0(t)) + \tilde{f}(h^0(v^0(t)), v^0(t)), \\
v^0(t) &= e^{tB} v_0 + \int_0^t e^{(t-s)B} g(h(v^0(s)), v^0(s)) ds.
\end{aligned}$$

By the existence results (Prop. 2.6 and Prop. 2.8) these equations are well-defined. In addition, estimating the last equation yields

$$\|v^0(t)\|_{Y_1} \leq M_B e^{\omega_B t} \|v_0\|_{Y_1} + C_B L_g \int_0^t \frac{e^{(t-s)\omega_B}}{(t-s)^{1-\delta}} (1 + L_{h^0}) \|v^0(s)\|_{Y_1} ds.$$

This can be further estimated by applying Gronwall's inequality

$$\|v^0(t)\|_{Y_1} \leq M_B e^{\omega_B t} e^{C_B L_g (1+L_{h^0})\Gamma(\delta)} \|v_0\|_{Y_1}. \tag{2.14}$$

In a similar fashion we obtain that $\partial_t v^0$ satisfies

$$\partial_t v^0(t) = e^{Bt} (Bv_0 + g(h^0(v_0), v_0)) + \int_0^t e^{B(t-s)} \partial_s g(h^0(v^0(s)), v^0(s)) ds$$

and we estimate

$$\begin{aligned} \|\partial_t v^0\|_Y &\leq M_B e^{\omega_B t} (\|Bv_0\|_Y + L_g(L_{h^0} + 1)\|v_0\|_Y) \\ &\quad + M_B \int_0^t e^{\omega_B(t-s)} \|Dg(h^0(v^0), v^0)\|_{\mathcal{B}(X_1 \times Y_1, Y)} \|\partial_s v^0\|_Y ds \end{aligned}$$

which yields, assuming that the total derivative of g is bounded by L_g ,

$$\|\partial_t v^0\|_Y \leq CM_B e^{\omega_B t} (\|Bv_0\|_Y + L_g(L_{h^0} + 1)\|v_0\|_Y). \quad (2.15)$$

We start with the estimates for the u component in the norm $\|\cdot\|_{X_\alpha}$, where $\alpha \in (0, 1)$ is chosen such that $0 \leq \gamma - (1 - \alpha) \leq 1$. The reason for the choice of norm becomes clear when looking at the details of the estimate. We apply the triangle inequality to split the estimate into three parts

$$\begin{aligned} \|u^\varepsilon(t) - h^0(v^0(t))\|_{X_\alpha} &\leq \underbrace{\|u^\varepsilon(t) - \tilde{u}^\varepsilon(t)\|_{X_\alpha}}_{\text{Prop. 2.11}} + \underbrace{\|\tilde{u}^\varepsilon(t) - u^{\varepsilon,0}(t)\|_{X_\alpha}}_{\text{Prop. 2.12}} \\ &\quad + \underbrace{\|u^{\varepsilon,0}(t) - h^0(v^0(t))\|_{X_\alpha}}_{\text{Prop. 2.13}}. \end{aligned}$$

Combining the three estimates for the u component (2.10)-(2.13) yields

$$\begin{aligned} \|u^\varepsilon(t) - h^0(v^0(t))\|_{X_\alpha} &\leq \\ &\leq C \sup_{0 \leq s \leq t} \|v^\varepsilon(s) - v^0(s)\|_{Y_1} + CM_A e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t} \|u_0 - h^0(v_0)\|_{X_\alpha} \quad (2.16) \\ &\quad + C\varepsilon^{1-\alpha} (e^{\omega_B t} + \varepsilon^{-\gamma} t^\gamma e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t}) \|v_0\|_{Y_1}, \end{aligned}$$

where the constants C are independent of ε and only depend on the nonlinearities f and g and on the operators A and B .

Now, for the equation in v we estimate

$$\begin{aligned} \|v^\varepsilon(t) - v^0(t)\|_{Y_1} &\leq \\ &\leq L_g C_B \int_0^t \frac{e^{(t-s)\omega_B}}{(t-s)^{1-\delta}} (\|u^\varepsilon(s) - h(v^0(s))\|_{X_1} + \|v^\varepsilon - v^0\|_{Y_1}) ds. \end{aligned}$$

Using the previous estimate (2.16) for u yields

$$\begin{aligned} \|v^\varepsilon(t) - v^0(t)\|_{Y_1} &\leq \\ &\leq L_g C_B \int_0^t \frac{e^{(t-s)\omega_B}}{(t-s)^{1-\delta}} \left(\|v^\varepsilon(s) - v^0(s)\|_{Y_1} + \sup_{0 \leq \tau \leq s} \|v^\varepsilon(\tau) - v^0(\tau)\|_{Y_1} \right. \\ &\quad + CM_A e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)s} \|u_0 - h^0(v_0)\|_{X_\alpha} \\ &\quad \left. + C\varepsilon^{1-\alpha} (e^{\omega_B s} + \varepsilon^{-\gamma} s^\gamma e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)s}) \|v_0\|_{Y_1} \right) ds. \end{aligned}$$

Since the right-hand side is increasing in t we have

$$\begin{aligned} \sup_{0 \leq s \leq t} \|v^\varepsilon(s) - v^0(s)\|_{Y_1} &\leq \\ &\leq C \int_0^t \frac{e^{(t-s)\omega_B}}{(t-s)^{1-\delta}} \left(\sup_{0 \leq \tau \leq s} \|v^\varepsilon(\tau) - v^0(\tau)\|_{Y_1} + C e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)s} \|u_0 - h^0(v_0)\|_{X_\alpha} \right. \end{aligned}$$

$$+ C\varepsilon^{1-\alpha}(e^{\omega_B s} + \varepsilon^{-\gamma} s^\gamma e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)s}) \|v_0\|_{Y_1} \Big) ds.$$

Applying Gronwall's inequality yields

$$\begin{aligned} \sup_{0 \leq s \leq t} \|v^\varepsilon(s) - v^0(s)\|_{Y_1} &\leq \\ &\leq C e^{e^{\omega_B t}} \int_0^t \frac{e^{(t-s)\omega_B}}{(t-s)^{1-\delta}} \left(C e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)s} \|u_0 - h^0(v_0)\|_{X_\alpha} \right. \\ &\quad \left. + C \varepsilon^{1-\alpha}(e^{\omega_B s} + \varepsilon^{-\gamma} s^\gamma e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)s}) \|v_0\|_{Y_1} \right) ds. \end{aligned}$$

This estimate can be further improved to

$$\begin{aligned} \sup_{0 \leq s \leq t} \|v^\varepsilon(s) - v^0(s)\|_{Y_1} &\leq \\ &\leq C e^{e^{\omega_B t} + ct\delta} \varepsilon^{1-\alpha} (e^{\omega_B t} + \varepsilon^{-\gamma} t^\gamma e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t}) \|v_0\|_{Y_1} \\ &\quad + C e^{e^{\omega_B t} + ct} C M_A e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t} \|u_0 - h^0(v_0)\|_{X_\alpha}. \end{aligned} \quad (2.17)$$

Hence, combining the estimate for the u and v component yields

$$\begin{aligned} \|u^\varepsilon(t) - h^0(v^0(t))\|_{X_\alpha} + \|v^\varepsilon(t) - v^0(t)\|_{Y_1} &\leq \\ &\leq C_1 e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t} \|u_0 - h^0(v_0)\|_{X_\alpha} \\ &\quad + C_2 \varepsilon^{1-\alpha} (e^{\omega_B t} + \varepsilon^{-\gamma} t^\gamma e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t}) \|v_0\|_{Y_1}, \end{aligned}$$

which is the desired inequality. $\square$

3. Slow manifolds

In this section we generalize the Fenichel theory for ODEs to infinite dimensional systems. This theory originates from the works of Fenichel as a perturbation theory of normally hyperbolic invariant manifolds [23] and provides a suitable framework for the treatment of fast-slow ODE systems [24, 40, 42].

Following the structure of the Fenichel theorem in finite dimension we establish the existence of a slow manifold, show that the distance between this slow manifold and the critical manifold is small in a suitable sense and that the slow flow on the critical manifold is a good approximation of the flow on the slow manifold.

One important aspect of the Fenichel theory is the notion of invariant manifold. A manifold M is forward invariant with respect to the semigroup $(T(t))_{t \geq 0}$ if for all $m \in M$ and all $t \geq 0$ it holds that $T(t)m \in M$.

Theorem 3.1. (*General Fenichel Theorem for infinite-dimensional fast-reaction systems*) Suppose $S_{0,\zeta}$ is a submanifold of the critical manifold S_0 and let $\tilde{A}_\varepsilon$ be the operator of the previous section introduced in system (2.5). In addition, let the technical assumptions of Sections 2.1 and 3.1 hold. Then, for $\varepsilon, \zeta > 0$ small enough the following hold

- (GF1) *There exists a locally forward invariant manifold S_ε for the system (3.2) called the slow manifold. Local invariance means that trajectories can enter or leave S_ε only through its boundaries;*
- (GF2) *The slow manifold S_ε has a distance of $\mathcal{O}(\varepsilon)$ to S_0 in the Y_1 -norm, i.e. the norm generated by the operator B ;*
- (GF3) *The slow manifold S_ε is C^1 -regular in the topology induced by the operators A and B ;*
- (GF4) *The flow on S_ε converges to the semi-flow on $S_{0,\zeta}$, where again the convergence is with respect to the Y_1 -norm.*

The details as well as the proofs of each of the statements follow in the subsequent subsections.

Remark 3.2. We want to remark on the difference between the results presented in this section and the result by Bates et al. in [2]. One key assumption in the later work is that

$$\left\| \begin{pmatrix} u^\varepsilon(t) - u^0(t) \\ v^\varepsilon(t) - v^0(t) \end{pmatrix} \right\|_{C^1(N, X \times Y)} \rightarrow 0 \quad \text{as } \varepsilon \rightarrow 0,$$

where N is a neighborhood of S_0 . When we compare this assumption with Theorem 2.14 we observe that we only require

$$\left\| \begin{pmatrix} u^\varepsilon(t) - u^0(t) \\ v^\varepsilon(t) - v^0(t) \end{pmatrix} \right\|_{C(N, X_1 \times Y_1)} \rightarrow 0 \quad \text{as } \varepsilon \rightarrow 0$$

for $t \in [0, \infty)$ to hold. Hence, our result applies to a larger class of fast-reaction systems. One way to satisfy the assumption in [2] is to assume that the operator B generates a C^0 -group. We, however, choose a different approach in extending the Fenichel theory to infinite dimensions. Lastly, in the case of the equation for the slow component is an ODE or if B generates a C^0 -group we observe that the results of Theorem 3.1 coincide with the ones in [2].

3.1. Assumptions

To show the results in this section we need additional assumptions on the operator B and the Banach space Y . The reason for this is that we want to express all the terms that arise in the upcoming estimates in terms of the slow variable v . However, when we rewrite the fast reaction system in the slow time scale by $\tau = \varepsilon t$

$$\begin{aligned} \partial_\tau u^\varepsilon &= \tilde{A}_\varepsilon u^\varepsilon + \tilde{f}(u^\varepsilon, v^\varepsilon), \\ \partial_\tau v^\varepsilon &= \varepsilon(Bv^\varepsilon + g(u^\varepsilon, v^\varepsilon)) \end{aligned}$$

we observe that in the limit $\varepsilon \rightarrow 0$ the expression $\varepsilon Bv^\varepsilon$ might not be well-defined as B is an unbounded operator. For more details and an example we refer to [35, Section 3.3].

Therefore, we introduce a splitting in the slow variable space

$$Y = Y_F^\zeta \oplus Y_S^\zeta$$

into a fast part Y_F^ζ and a slow part Y_S^ζ , where $\zeta > 0$ is a small parameter, such that

Assumptions on the space Y and the operator B :

- (Y1) The spaces Y_F^ζ and Y_S^ζ are closed in Y and the projections $\text{pr}_{Y_F^\zeta}$ and $\text{pr}_{Y_S^\zeta}$ commute with B on Y_1 .
- (Y2) The spaces $Y_F^\zeta \cap Y_1$ and $Y_S^\zeta \cap Y_1$ are closed subspaces of Y_1 and are endowed with the norm $\|\cdot\|_{Y_1}$.
- (Y3) The realization of B in Y_F^ζ , i.e.

$$B_{Y_F^\zeta} : D(Y_F^\zeta) \subset Y_F^\zeta \rightarrow Y_F^\zeta, \quad v \mapsto Bv$$

with

$$D(Y_F^\zeta) := \{v_0 \in Y_F^\zeta : Bv_0 \in Y_F^\zeta\}$$

has 0 in its resolvent set.

- (Y4) The realization of the operator B in the slow subspace Y_S^ζ generates a C_0 -group $(e^{tB_{Y_S^\zeta}})_{t \in \mathbb{R}} \subset \mathcal{B}((Y_S^\zeta, \|\cdot\|_Y))$ which for $t \geq 0$ satisfies $e^{tB_{Y_S^\zeta}} = e^{tB}$ on Y_S^ζ .
- (Y5) The fast subspace Y_F^ζ contains the parts of Y_1 that decay under the semigroup $(e^{tB})_{t \geq 0}$ almost as fast as functions in the fast variable space X_1 under the semi-group $(e^{\zeta^{-1}t\tilde{A}_\varepsilon})_{t \geq 0}$ generated by the modified operator $\tilde{A}_\varepsilon$. The space Y_S^ζ on the other hand contains the parts of Y_1 that do not decay or which decay only slowly under the semigroup $(e^{tB})_{t \geq 0}$ compared to X_1 under $(e^{\zeta^{-1}t\tilde{A}_\varepsilon})_{t \geq 0}$. Hence, there are constants $C_B, M_B > 0$ such that for $\zeta > 0$ small enough there are constants N_F^ζ, N_S^ζ satisfying

$$0 \leq N_F^\zeta < N_S^\zeta < |\zeta^{-1}(\varepsilon\omega_A + \lambda_0)|$$

such that for all $t \geq 0$ and $y_F \in Y_F^\zeta, y_S \in Y_S^\zeta$ we have

$$\begin{aligned} \|e^{tB}y_F\|_{Y_1} &\leq C_B t^{\delta-1} e^{(N_F^\zeta + \zeta^{-1}(\varepsilon\omega_A + \lambda_0))t} \|y_F\|_{Y_\delta}, \\ \|e^{-tB}y_S\|_{Y_1} &\leq M_B e^{-(N_S^\zeta + \zeta^{-1}(\varepsilon\omega_A + \lambda_0))t} \|y_S\|_{Y_1}. \end{aligned}$$

- (Y6) A sufficient condition on the size of the parameter $\zeta > 0$ is the relation $\varepsilon\zeta^{-1} \leq c < 1$. This assures that the following inequality holds true

$$(1 - \varepsilon\zeta^{-1})(\varepsilon\omega_A + \lambda_0) - \frac{\varepsilon}{2}(N_S^\zeta + N_F^\zeta) < 0.$$

- (Y7) Lastly, the following spectral gap condition holds $L < 1$, where

$$L := \frac{2^\gamma L_f C_A \Gamma(\gamma)}{(2(\varepsilon\zeta^{-1} - 1)(\varepsilon\omega_A + \lambda_0) + \varepsilon(N_S^\zeta + N_F^\zeta))^\gamma} + \frac{2^\delta L_g C_B \Gamma(\delta)}{(N_S^\zeta - N_F^\zeta)^\delta} + \frac{2L_g M_B \Gamma(\delta)}{N_S^\zeta - N_F^\zeta} \quad (3.1)$$

Remark 3.3. Here, we take a closer look at the terms arising in the spectral gap condition (3.1). As $\varepsilon\zeta^{-1} \leq c < 1$ holds we have $(1 - \varepsilon\zeta^{-1})(\varepsilon\omega_A + \lambda_0) > 0$. Thus, the first term is independent of ε and ζ and can always be controlled by a constant $\tilde{C} < 1$.

The quantities N_S^ζ , N_F^ζ can be seen as shifted eigenvalues of the operator $\tilde{A}_\varepsilon$ and therefore $N_S^\zeta - N_F^\zeta$ can be considered as the spectral gap of this operator. It relates to the parameter ζ through $N_S^\zeta - N_F^\zeta = C(\zeta)$, where $C(\zeta) \rightarrow \infty$ as $\zeta \rightarrow 0$, (see Subsection 3.5 for an example).

However, this spectral gap condition may be satisfied in low dimensions only. Recalling that the Weyl asymptotic for the eigenvalues of the Laplacian reads $\lambda_m \sim Cm^{2/d}$ yields for a domain $\Omega \subset \mathbb{R}$ that $N_S^\zeta - N_F^\zeta = \mathcal{O}(\zeta^{-1/2})$. For a general domain $\Omega \subset \mathbb{R}^d$, $d \geq 2$ this asymptotic may fail. Yet, for special domains such as $\Omega = [0, 2\pi]^2$ a number theoretical result from [50] yields that $N_S^\zeta - N_F^\zeta = \mathcal{O}(\log(\zeta))$. Methods to overcome this problem are discussed in [41, 46, 57].

With this splitting we can rewrite the fast reaction system in the following way

$$\begin{aligned} \partial_t u^\varepsilon(t) &= \frac{1}{\varepsilon} \tilde{A}_\varepsilon u^\varepsilon(t) + \tilde{f}(u^\varepsilon(t), v_F^\varepsilon(t), v_S^\varepsilon(t)), \\ \partial_t v_F^\varepsilon(t) &= B v_F^\varepsilon(t) + \text{pr}_{Y_F^\zeta} g(u^\varepsilon(t), v_F^\varepsilon(t), v_S^\varepsilon(t)), \\ \partial_t v_S^\varepsilon(t) &= B v_S^\varepsilon(t) + \text{pr}_{Y_S^\zeta} g(u^\varepsilon(t), v_F^\varepsilon(t), v_S^\varepsilon(t)), \\ u^\varepsilon(0) &= u_0, \quad v_F^\varepsilon(0) = \text{pr}_{Y_F^\zeta} v_0, \quad v_S^\varepsilon(0) = \text{pr}_{Y_S^\zeta} v_0. \end{aligned} \quad (3.2)$$

3.2. Existence of slow manifolds

We construct a family of slow manifolds $S_{\varepsilon, \zeta}$ as

$$S_{\varepsilon, \zeta} := \{(h^{\varepsilon, \zeta}(v_0), v_0) : v_0 \in Y_1 \cap Y_S^\zeta\}, \quad (3.3)$$

where the function $h^{\varepsilon, \zeta}$ maps

$$h^{\varepsilon, \zeta} : (Y_S^\zeta \cap Y_1) \rightarrow X_1 \times (Y_F^\zeta \cap Y_1).$$

The next step is to construct the functions $h^{\varepsilon, \zeta}$ using a Lyapunov-Perron method (cf. [8, 31]). To this end, we introduce the operator

$$\mathcal{L}_{v_0, \varepsilon, \zeta} : C_\eta \rightarrow C_\eta,$$

$$\begin{pmatrix} u \\ v_F \\ v_S \end{pmatrix} \mapsto \left[t \mapsto \begin{pmatrix} \varepsilon^{-1} \int_{-\infty}^t e^{(t-s)\tilde{A}_\varepsilon} \tilde{f}(u(s), v_F(s), v_S(s)) ds \\ \int_{-\infty}^t e^{(t-s)B} \text{pr}_{Y_F^\zeta} g(u(s), v_F(s), v_S(s)) ds \\ e^{tB} v_0 + \int_0^t e^{(t-s)B} \text{pr}_{X_S^\zeta} g(u(s), v_F(s), v_S(s)) ds \end{pmatrix} \right], \quad (3.4)$$

where $v_0 \in Y_1 \cap Y_S^\zeta$ and

$$C_\eta := C((-\infty, 0], e^{\eta t}; X_1 \times (Y_F^\zeta \cap Y_1) \times (Y_1 \cap Y_S^\zeta))$$

is the space of $(u, v_F, v_S) \in C((-\infty, 0]; X_1 \times (Y_F^\zeta \cap Y_1) \times (Y_1 \cap Y_S^\zeta))$ such that

$$\|(u, v_F, v_S)\|_{C_\eta} := \sup_{t \leq 0} e^{-\eta t} (\|u\|_{X_1} + \|v_F\|_{Y_1} + \|v_S\|_{Y_1}) < \infty.$$

Proposition 3.4. *Let $v_0 \in Y_1 \cap Y_S^\zeta$. Then the operator $\mathcal{L}_{v_0, \varepsilon, \zeta}$ has a unique fixed point in C_η .*

Proof. As a first step we rewrite the system into the modified fast-slow system while maintaining the splitting in the slow variable v , where we replace the operator A by $\frac{1}{\varepsilon}\tilde{A}_\varepsilon$ and the nonlinear function f by $\tilde{f}$. Then, we show that the Lyapunov-Perron operator introduced in equation (3.4) has a unique fixed point. To do this, we apply the existence result shown in [35, Prop. 5.1], where we set

$$\eta = \zeta^{-1}(\varepsilon\omega_A + \lambda_0) + \frac{N_S^\zeta + N_F^\zeta}{2}.$$

□

The function $h^{\varepsilon,\zeta}$ used to construct the family of slow manifolds is given by the first two components of the fixed point of the Lyapunov-Perron operator $\mathcal{L}_{v_0,\varepsilon,\zeta}$ evaluated at $t = 0$, i.e. $h^{\varepsilon,\zeta}(v_0) = (u^{v_0}(0), v_F^{v_0}(0))$. This completes the first part of the statement (GF1) in Theorem 3.1. In addition, the invariance of the slow manifold follows from the construction of the Lyapunov-Perron operator, where we consider all solutions in backward time that can develop from the initial condition in the slow component.

Proposition 3.5. *Let $(u^{v_0}, v_F^{v_0}, v_S^{v_0})$ be the unique fixed point of $\mathcal{L}_{v_0,\varepsilon,\zeta}$ evaluated at $t = 0$ and assume that*

$$\frac{2^\gamma L_{\tilde{f}} C_A \Gamma(\gamma)}{(2(\varepsilon\zeta^{-1} - 1)(\varepsilon\omega_A + \lambda_0) + \varepsilon(N_S^\zeta + N_F^\zeta))^\gamma} + \frac{2^\delta L_g C_B \Gamma(\delta)}{(N_S^\zeta - N_F^\zeta)^\delta} + \frac{2L_g M_B \Gamma(\delta)}{N_S^\zeta - N_F^\zeta} < 1.$$

Then, the mapping

$$h^{\varepsilon,\zeta} : (Y_S^\zeta \cap Y_1) \rightarrow X_1 \times (Y_F^\zeta \cap Y_1), v_0 \mapsto (u^{v_0}(0), v_F^{v_0}(0))$$

is Lipschitz continuous with Lipschitz constant $L_\zeta = M_B/(1 - L)$.

Proof. This follows from the estimates in Proposition 3.4, cf. [35, Prop 5.2].

□

3.3. Distance to critical manifold

Proposition 3.6. *There exists a constant $C > 0$ depending on the operator A and the nonlinear functions f and g such that for all $\varepsilon, \zeta > 0$ small enough satisfying $\varepsilon\zeta^{-1} < 1$ and all initial data $v_0 \in Y_S^\zeta$ it holds that*

$$\left\| \begin{pmatrix} h_{X_1}^{\varepsilon,\zeta}(v_0) - h^0(v_0) \\ h_{Y_F}^{\varepsilon,\zeta}(v_0) \end{pmatrix} \right\|_{X_\alpha \times Y_1} \leq C \left(\varepsilon^{1-\alpha} + \frac{1}{(N_S^\zeta - N_F^\zeta)^\delta} \right) \|v_0\|_{Y_1}. \quad (3.5)$$

Proof. Let $(\bar{u}, \bar{v}_F, \bar{v}_S) \in C_\eta$ be the unique fixed point of $\mathcal{L}_{v_0,\varepsilon,\zeta}$. That is we have

$$(\bar{u}, \bar{v}_F, \bar{v}_S) = (h_{X_1}^{\varepsilon,\zeta}(\bar{v}_S), h_{Y_F}^{\varepsilon,\zeta}(\bar{v}_S), \bar{v}_S).$$

Since $(\bar{u}, \bar{v}_F, \bar{v}_S)$ solves the system (3.2) on $(-\infty, 0]$ we have that $\bar{v}_S \in C^1((-\infty, 0], e^{\eta t}; Y)$. We estimate that

$$\|h_{Y_F}^{\varepsilon,\zeta}(\bar{v}_S(t))\|_{Y_1} =$$

$$\begin{aligned}
&= \left\| \int_{-\infty}^t e^{(t-s)B} \operatorname{pr}_{Y_F^\zeta} g(h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)), h_{Y_F}^{\varepsilon, \zeta}(\bar{v}_S(s)), \bar{v}_S(s)) \, ds \right\|_{Y_1} \\
&\leq L_g C_B e^{\eta t} \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)), h_{Y_F}^{\varepsilon, \zeta}(\bar{v}_S(s)), \bar{v}_S(s)\|_{C_\eta} \int_{-\infty}^t \frac{e^{(t-s)(\zeta^{-1}\omega_A + N_F^\zeta - \eta)}}{(t-s)^{1-\delta}} \, ds \\
&\leq \frac{L_\zeta L_g C_B \Gamma(\delta) e^{\eta t}}{(\eta - \zeta^{-1}(\varepsilon\omega_A + \lambda_0) - N_F^\zeta)^\delta} \|v_0\|_{Y_1} \\
&\leq C \frac{1}{N_S^\zeta - N_F^\zeta} \|v_0\|_{Y_1}.
\end{aligned}$$

In addition, for $t_0 \leq t \leq 0$ we compute

$$\begin{aligned}
&h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(t)) - h^0(\bar{v}_S(t)) = \\
&= \varepsilon^{-1} \int_{-\infty}^t e^{(t-s)\varepsilon^{-1}\tilde{A}_\varepsilon} \tilde{f}(h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)), h_{Y_F}^{\varepsilon, \zeta}(\bar{v}_S(s)), \bar{v}_S(s)) \, ds - h^0(\bar{v}_S(t)) \\
&= \varepsilon^{-1} \int_{-\infty}^t e^{(t-s)\varepsilon^{-1}\tilde{A}_\varepsilon} \tilde{f}(h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)), h_{Y_F}^{\varepsilon, \zeta}(\bar{v}_S(s)), \bar{v}_S(s)) \, ds \\
&\quad + (D_u f(h^0(v^0), v^0))^{-1} \tilde{f}(h^0(\bar{v}_S), 0, \bar{v}_S) - (\tilde{A}_\varepsilon)^{-1} \tilde{f}(h^0(\bar{v}_S), 0, \bar{v}_S) \\
&\quad + (\tilde{A}_\varepsilon)^{-1} \tilde{f}(h^0(\bar{v}_S), 0, \bar{v}_S)
\end{aligned}$$

Using integration by parts yields

$$\begin{aligned}
&= \varepsilon^{-1} \int_{-\infty}^{t_0} e^{(t-s)\varepsilon^{-1}\tilde{A}_\varepsilon} \tilde{f}(h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)), h_{Y_F}^{\varepsilon, \zeta}(\bar{v}_S(s)), \bar{v}_S(s)) \, ds \\
&\quad + \int_{t_0}^t e^{(t-s)\varepsilon^{-1}\tilde{A}_\varepsilon} (\tilde{A}_\varepsilon)^{-1} \partial_s \tilde{f}(h^0(\bar{v}_S(s)), 0, \bar{v}_S(s)) \, ds \\
&\quad + \varepsilon^{-1} \int_{t_0}^t e^{(t-s)\varepsilon^{-1}\tilde{A}_\varepsilon} \left(\tilde{f}(h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)), h_{Y_F}^{\varepsilon, \zeta}(\bar{v}_S(s)), \bar{v}_S(s)) \right. \\
&\quad \left. - \tilde{f}(h^0(\bar{v}_S(s)), 0, \bar{v}_S(s)) \right) \, ds \\
&\quad + e^{(t-t_0)\varepsilon^{-1}\tilde{A}_\varepsilon} (\tilde{A}_\varepsilon)^{-1} \tilde{f}(h^0(\bar{v}_S(t_0)), 0, \bar{v}_S(t_0)) \\
&\quad + ((D_u f(h^0(v^0), v^0))^{-1} - (\tilde{A}_\varepsilon)^{-1}) \tilde{f}(h^0(\bar{v}_S), 0, \bar{v}_S).
\end{aligned}$$

Again we observe the additional terms in the last line due to the modified operator $\tilde{A}_\varepsilon$, which will lead again to a lower regularity of the estimate, when comparing this proof with the one in [35]. Hence, we estimate the above expression as follows

$$\begin{aligned}
& \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S) - h^0(\bar{v}_S)\|_{X_\alpha} \leq \\
& \leq C_A L_{\tilde{f}} \int_{-\infty}^{t_0} \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)), h_{Y_F}^{\varepsilon, \zeta}(\bar{v}_S(s)), \bar{v}_S(s)\|_{X_1 \times Y_1 \times Y_1} ds \\
& + \varepsilon \|(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X, X_{\gamma-(1-\alpha)})} \int_{t_0}^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} ds \\
& \times \|\tilde{f}(h^0(\bar{v}_S), \bar{v}_S)\|_{L^\infty((-\infty, 0]; X)} \\
& + C_A L_{\tilde{f}} \int_{t_0}^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)) - h^0(\bar{v}_S(s)), h_{Y_F}^{\varepsilon, \zeta}(\bar{v}_S(s))\|_{X_\alpha \times Y_1} ds \\
& + M_A L_{\tilde{f}} e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)(t-t_0)} \|(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X_\gamma, X_1)} \|h^0(\bar{v}_S(t_0)), \bar{v}_S(t_0)\|_{X_1 \times Y_1} \\
& + L_{\tilde{f}} \varepsilon \|(\mathcal{D}_u f(h^0(v^0), v^0))^{-1} J\|_{\mathcal{B}(X_{\gamma-(1-\alpha)}, X_1)} \|h^0(\bar{v}_S(t)), \bar{v}_S(t)\|_{X_1 \times Y_1} \\
& \leq C_A L_{\tilde{f}} e^{-(t-t_0)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0) + \eta t_0} \int_{-\infty}^{t_0} \frac{e^{(t_0-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0 - \varepsilon\eta)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} ds \\
& \times \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S), h_{Y_F}^{\varepsilon, \zeta}(\bar{v}_S), \bar{v}_S\|_{C_\eta} \\
& + \varepsilon L_{\tilde{f}} \|(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X, X_{\gamma-(1-\alpha)})} \int_{t_0}^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} ds \times \\
& \times (1 + L_{h^0}) \tilde{C} e^{\omega_B t} (\|Bv_0\|_Y + L_g(L_{h^0} + 1)\|v_0\|_Y + \|v_0\|_{Y_1}) \\
& + C_A L_{\tilde{f}} \int_{t_0}^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)) - h^0(\bar{v}_S(s))\|_{X_\alpha} ds \\
& + C_A L_{\tilde{f}} \frac{L_\zeta L_g C_B \Gamma(\delta)}{(\eta - \zeta^{-1}(\varepsilon\omega_A + \lambda_0) - N_F^\zeta)^\delta} \int_{t_0}^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} e^{\eta s} ds \|v_0\|_{Y_1} \\
& + M_A L_{\tilde{f}} e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)(t-t_0) + \eta t_0} \|(\tilde{A}_\varepsilon)^{-1}\|_{\mathcal{B}(X_\gamma, X_1)} \|h^0(\bar{v}_S), 0, \bar{v}_S\|_{C_\eta} \\
& + L_{\tilde{f}} \lambda_0^{-1} \varepsilon \|J\|_{\mathcal{B}(X_\alpha, X_1)} (L_{h^0} + 1) \|\bar{v}_S(t)\|_{Y_1}.
\end{aligned}$$

Using the Lipschitz continuity of $h^{\varepsilon, \zeta}$ and the observation of Lemma 2.10 yields

$$\begin{aligned}
& \leq C_A L_{\tilde{f}} \frac{e^{t_0\varepsilon^{-1}(\varepsilon\eta - (\varepsilon\omega_A + \lambda_0))} e^{t\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{-\omega_A + \varepsilon^{-1}\lambda_0 + \eta} L_\zeta \|v_0\|_{Y_1} \\
& + \varepsilon^{2-\gamma-\alpha} C L_{\tilde{f}} \frac{\Gamma(\gamma)}{|\varepsilon\omega_A - \lambda_0|^\gamma} (1 + L_{h^0}) \tilde{C} e^{\omega_B t} \\
& \times (\|Bv_0\|_Y + L_g(L_{h^0} + 1)\|v_0\|_Y + \|v_0\|_{Y_1}) \\
& + C_A L_{\tilde{f}} \frac{L_\zeta L_g C_B \Gamma(\delta)}{(\eta - \zeta^{-1}(\varepsilon\omega_A + \lambda_0) - N_F^\zeta)^\delta} \frac{\Gamma(\gamma)}{|\varepsilon\omega_A - \lambda_0|^\gamma} e^{\eta t} \|v_0\|_{Y_1} \\
& + M_A L_{\tilde{f}} L_\zeta \varepsilon^{\gamma-1} C e^{t_0\varepsilon^{-1}(\varepsilon\eta - (\varepsilon\omega_A + \lambda_0))} e^{t\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)} \|v_0\|_{Y_1} \\
& + L_{\tilde{f}} \lambda_0^{-1} \varepsilon^{1-\alpha} C (L_{h^0} + 1) M_B e^{\omega_B t} e^{C_B L_g(1+L_{h^0})\Gamma(\delta)} \|v_0\|_{Y_1} \\
& + C_A L_{\tilde{f}} \int_{t_0}^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)) - h^0(\bar{v}_S(s))\|_{X_\alpha} ds.
\end{aligned}$$

We observe that the right-hand side of the inequality is of the form

$$\begin{aligned} & \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S) - h^0(\bar{v}_S)\|_{X_\alpha} \leq \\ & \leq R(t) + C_A L_{\tilde{f}} \int_{t_0}^t \frac{e^{(t-s)\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)}}{\varepsilon^\gamma(t-s)^{1-\gamma}} \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(s)) - h^0(\bar{v}_S(s))\|_{X_\alpha} ds, \end{aligned}$$

where $R(t)$ is non-decreasing in time and thus

$$\begin{aligned} & \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(t)) - h^0(\bar{v}_S(t))\|_{X_\alpha} \leq \\ & \leq C_A L_{\tilde{f}} \frac{e^{t_0\varepsilon^{-1}(\varepsilon\eta - (\varepsilon\omega_A + \lambda_0))}}{-\omega_A + \varepsilon^{-1}\lambda_0 + \eta} L_\zeta \|v_0\|_{Y_1} \\ & + \varepsilon^{2-\gamma-\alpha} C L_{\tilde{f}} \frac{\Gamma(\gamma)(1+L_{h^0})}{|\varepsilon\omega_A - \lambda_0|^\gamma} (\|Bv_0\|_Y + L_g(L_{h^0} + 1)\|v_0\|_Y + \|v_0\|_{Y_1}) \\ & + C_A L_{\tilde{f}} \frac{L_\zeta L_g C_B \Gamma(\delta)}{(\eta - \zeta^{-1}(\varepsilon\omega_A + \lambda_0) - N_F^\zeta)^\delta} \frac{\Gamma(\gamma)}{|\varepsilon\omega_A - \lambda_0|^\gamma} \|v_0\|_{Y_1} \\ & + M_A L_{\tilde{f}} L_\zeta \varepsilon^{\gamma-1} e^{t_0\varepsilon^{-1}(\varepsilon\eta - (\varepsilon\omega_A + \lambda_0))} \|v_0\|_{Y_1} \\ & + L_{\tilde{f}} \lambda_0^{-1} \varepsilon^{1-\alpha} C(L_{h^0} + 1) M_B e^{C_B L_g(1+L_{h^0})\Gamma(\delta)} \|v_0\|_{Y_1}. \end{aligned}$$

Using the assumption that $\varepsilon\zeta^{-1} < 1$ we have that $\varepsilon\eta - (\varepsilon\omega_A + \lambda_0) > 0$. Hence, we can let $t_0 \rightarrow -\infty$ and obtain

$$\begin{aligned} & C_1^{-1} \|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(t)) - h^0(\bar{v}_S(t))\|_{X_1} \leq \\ & \leq \varepsilon^{2-\gamma-\alpha} L_{\tilde{f}} C \frac{\Gamma(\gamma)}{|\varepsilon\omega_A - \lambda_0|^\gamma} (1 + L_{h^0}) \tilde{C} (\|Bv_0\|_Y + L_g(L_{h^0} + 1)\|v_0\|_Y + \|v_0\|_{Y_1}) \\ & + C_A L_{\tilde{f}} \frac{L_\zeta L_g C_B \Gamma(\delta)}{(\eta - \zeta^{-1}(\varepsilon\omega_A + \lambda_0) - N_F^\zeta)^\delta} \frac{\Gamma(\gamma)}{|\varepsilon\omega_A - \lambda_0|^\gamma} \|v_0\|_{Y_1} \\ & + L_{\tilde{f}} \lambda_0^{-1} \varepsilon^{1-\alpha} C(L_{h^0} + 1) M_B e^{C_B L_g(1+L_{h^0})\Gamma(\delta)} \|v_0\|_{Y_1}. \end{aligned}$$

Focusing on the terms in ε and using $\eta = \zeta^{-1}(\varepsilon\omega_A + \lambda_0) + \frac{N_F^\zeta + N_S^\zeta}{2}$ yields the desired estimate

$$\|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(t)) - h^0(\bar{v}_S(t))\|_{X_\alpha} \leq C \left(\varepsilon^{1-\alpha} + \frac{1}{(N_S^\zeta - N_F^\zeta)^\delta} \right) \|v_0\|_{Y_1}.$$

This shows statement (GF2) of the generalized Fenichel Theorem 3.1. $\square$

Remark 3.7. If we have additional knowledge on the convergence of $N_S^\zeta - N_F^\zeta$ we can improve the above result. Assuming that we have the asymptotic $N_S^\zeta - N_F^\zeta = C\zeta^{-\beta}$, for some $\beta > 0$ then we obtain for $\varepsilon > 0$ small enough

$$\|h_{X_1}^{\varepsilon, \zeta}(\bar{v}_S(t)) - h^0(\bar{v}_S(t))\|_{X_\alpha} \leq C \left(\min\{\varepsilon^{1-\gamma}, \varepsilon^{\alpha-1}\} + \zeta^\beta \right) \|v_0\|_{Y_1},$$

where we used the condition $\varepsilon\zeta^{-1} < 1$; for more background on the importance of double limits in singular perturbations we also refer to [43].

3.4. Attraction of trajectories and approximation by the slow flow

One technical result is to show the differentiability of the slow manifold. Here, we use the assumption that the nonlinearities f and g are continuously differentiable such that

$$\|Df(u, v)\|_{\mathcal{B}(X_1 \times Y_1, X_\gamma)} \leq L_f, \quad \|Dg(u, v)\|_{\mathcal{B}(X_1 \times Y_1, Y_\delta)} \leq L_g.$$

Proposition 3.8. *Under the previous assumptions in this section the slow manifold is differentiable, i.e. the mapping*

$$(Y_S^\zeta, \|\cdot\|_{Y_1}) \rightarrow (X_1, \|\cdot\|_{X_1}) \times (Y_F^\zeta, \|\cdot\|_{Y_1}), \quad v_0 \mapsto (h_{X_1}^{\varepsilon, \zeta}(v_0), h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0))$$

is differentiable.

Proof. The proof can be found in [35, Prop 5.4] where the key idea is to show that the derivative exists as the best local linear approximation of the graph of the manifold. $\square$

This result together with Proposition 3.6 yields the proof of statement (GF3) in Theorem 3.1. Moreover, this will allow us to prove the attraction of trajectories and the approximation by the slow flow which is the goal of the reminder of the section.

Next, let $(h_{X_1}^{\varepsilon, \zeta}(v_0), h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0), v_0) \in S_{\varepsilon, \zeta}$. Moreover, let (u, v_F, v_S) be the strict solution of (3.2) with initial data $(h_{X_1}^{\varepsilon, \zeta}(v_0), h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0), v_0)$, i.e. it holds that

$$\partial_t u(t) = \frac{1}{\varepsilon} \tilde{A}_\varepsilon u(t) + \frac{1}{\varepsilon} \tilde{f}(u(t), v_F(t), v_S(t))$$

and let $(u^\varepsilon, v_F^\varepsilon, v_S^\varepsilon)$ be the strict solution to system (3.2) with initial data $(u_0, v_{0,F}, v_{0,S})$. From the invariance and differentiability of the slow manifold it follows that $u(t) = h_{X_1}^{\varepsilon, \zeta}(v_S(t))$ and therefore

$$\begin{aligned} \partial_t u(t) &= \partial_t h_{X_1}^{\varepsilon, \zeta}(v_S(t)) = Dh_{X_1}^{\varepsilon, \zeta}(v_S(t))[\partial_t v_S(t)] \\ &= Dh_{X_1}^{\varepsilon, \zeta}(v_S(t))[Bv_S(t) + \text{pr}_{Y_S^\zeta} g(u(t), v_F(t), v_S(t))]. \end{aligned}$$

Combining both equations for $t = 0$ yields that

$$\begin{aligned} h_{X_1}^{\varepsilon, \zeta}(v_0) &= \varepsilon(\tilde{A}_\varepsilon)^{-1} Dh_{X_1}^{\varepsilon, \zeta}(v_0)[Bv_0 + \text{pr}_{Y_S^\zeta} g(h_{X_1}^{\varepsilon, \zeta}(v_0), h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0), v_0)] \\ &\quad - (\tilde{A}_\varepsilon)^{-1} f(h_{X_1}^{\varepsilon, \zeta}(v_0), h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0), v_0). \end{aligned} \quad (3.6)$$

Similarly we obtain

$$\begin{aligned} h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0) &= B_{Y_F^\zeta}^{-1} Dh_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0)[Bv_0 + \text{pr}_{Y_S^\zeta} g(h_{X_1}^{\varepsilon, \zeta}(v_0), h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0), v_0)] \\ &\quad - B_{Y_F^\zeta}^{-1} \text{pr}_{Y_F^\zeta} g(h_{X_1}^{\varepsilon, \zeta}(v_0), h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0), v_0). \end{aligned} \quad (3.7)$$

We note that these two relations hold for arbitrary $v_0 \in Y_S^\zeta$. Therefore it also holds for $v_0 = v_S^\varepsilon(t)$ for fixed $t \geq 0$ and we have

$$\partial_t h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(t)) = Dh_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(t))[Bv_S^\varepsilon(t) + \text{pr}_{Y_S^\zeta} g(u^\varepsilon(t), v_F^\varepsilon(t), v_S^\varepsilon(t))], \quad (3.8)$$

$$\partial_t h_{Y_F}^{\varepsilon, \zeta}(v_S^\varepsilon(t)) = Dh_{Y_F}^{\varepsilon, \zeta}(v_S^\varepsilon(t)) [Bv_S^\varepsilon(t) + \text{pr}_{Y_S^\zeta} g(u^\varepsilon(t), v_F^\varepsilon(t), v_S^\varepsilon(t))]. \quad (3.9)$$

Thus, we can express the solution as a flow on the slow manifold.

The next step in the generalization of the Fenichel theorem is to show that the fast components of the fast-reaction system approach the solutions on the slow manifold with an exponential rate in time.

Proposition 3.9. *For ε and ζ small enough satisfying $\varepsilon\zeta^{-1} < 1$ there exist constants $C, c > 0$ such that*

$$\left\| \begin{pmatrix} u^\varepsilon(t) - h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(t)) \\ v_F^\varepsilon(t) - h_{Y_F}^{\varepsilon, \zeta}(v_S^\varepsilon(t)) \end{pmatrix} \right\|_{X_1 \times Y_1} \leq Ce^{-ct} \left\| \begin{pmatrix} u_0 - h_{X_1}^{\varepsilon, \zeta}(v_{0,S}) \\ v_{0,F} - h_{Y_1}^{\varepsilon, \zeta}(v_{0,S}) \end{pmatrix} \right\|_{X_1 \times Y_1}.$$

Proof. Starting with the first component we compute

$$\begin{aligned} u^\varepsilon(t) - h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(t)) &= \\ &= e^{\varepsilon^{-1}t\tilde{A}_\varepsilon}(u_0 - h_{X_1}^{\varepsilon, \zeta}(v_{0,S})) + e^{\varepsilon^{-1}t\tilde{A}_\varepsilon}h_{X_1}^{\varepsilon, \zeta}(v_{0,S}) - h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(t)) \\ &\quad + \varepsilon^{-1} \int_0^t e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon} \tilde{f}(u^\varepsilon(s), v_F^\varepsilon(s), v_S^\varepsilon(s)) \, ds \\ &= e^{\varepsilon^{-1}t\tilde{A}_\varepsilon}(u_0 - h_{X_1}^{\varepsilon, \zeta}(v_{0,S})) - \int_0^t \partial_s(e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon}h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(s))) \, ds \\ &\quad + \varepsilon^{-1} \int_0^t e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon} \tilde{f}(u^\varepsilon(s), v_F^\varepsilon(s), v_S^\varepsilon(s)) \, ds \\ &= e^{\varepsilon^{-1}t\tilde{A}_\varepsilon}(u_0 - h_{X_1}^{\varepsilon, \zeta}(v_{0,S})) + \varepsilon^{-1} \int_0^t e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon} \tilde{A}_\varepsilon h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(s)) \, ds \\ &\quad - \int_0^t e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon} \partial_s h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(s)) \, ds \\ &\quad + \varepsilon^{-1} \int_0^t e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon} \tilde{f}(u^\varepsilon(s), v_F^\varepsilon(s), v_S^\varepsilon(s)) \, ds. \end{aligned}$$

Applying relations (3.6) and (3.8) yields

$$\begin{aligned} &= e^{\varepsilon^{-1}t\tilde{A}_\varepsilon}(u_0 - h_{X_1}^{\varepsilon, \zeta}(v_{0,S})) \\ &\quad + \int_0^t e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon} Dh_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(s)) \text{pr}_{Y_S^\zeta} \left[g(u^\varepsilon, v_F^\varepsilon, v_S^\varepsilon) \right. \\ &\quad \left. - g(h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(s)), h_{Y_F}^{\varepsilon, \zeta}(v_S^\varepsilon(s)), v_S^\varepsilon(s)) \right] \, ds \\ &\quad + \varepsilon^{-1} \int_0^t e^{\varepsilon^{-1}(t-s)\tilde{A}_\varepsilon} [f(u^\varepsilon, v_F^\varepsilon, v_S^\varepsilon) - f(h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(s)), h_{Y_F}^{\varepsilon, \zeta}(v_S^\varepsilon(s)), v_S^\varepsilon(s))] \, ds. \end{aligned}$$

Similarly, we obtain for the second component

$$\begin{aligned} v_F^\varepsilon(t) - h_{Y_F}^{\varepsilon, \zeta}(v_S^\varepsilon(t)) &= \\ &= e^{tB}(v_{0,F} - h_{Y_F}^{\varepsilon, \zeta}(v_{0,S})) \end{aligned}$$

$$\begin{aligned}
& + \int_0^t e^{(t-s)B} D h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_S^\varepsilon(s)) \operatorname{pr}_{Y_S^\zeta} \left[g(u^\varepsilon, v_F^\varepsilon, v_S^\varepsilon) \right. \\
& \quad \left. - g(h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(s)), h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_S^\varepsilon(s)), v_S^\varepsilon(s)) \right] ds \\
& + \int_0^t e^{(t-s)B} \operatorname{pr}_{Y_F^\zeta} [g(u^\varepsilon, v_F^\varepsilon, v_S^\varepsilon) - g(h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(s)), h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_S^\varepsilon(s)), v_S^\varepsilon(s))] ds.
\end{aligned}$$

Setting $z(t) = \|u^\varepsilon(t) - h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(t))\|_{X_1} + \|v_F^\varepsilon(t) - h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_S^\varepsilon(t))\|_{Y_1}$ we can estimate the two components as follows

$$\begin{aligned}
z(t) & \leq M_A e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)} \|u^\varepsilon(t) - h_{X_1}^{\varepsilon, \zeta}(v_S^\varepsilon(t))\|_{X_1} \\
& \quad + M_B e^{(\zeta^{-1}(\varepsilon\omega_A + \lambda_0) + N_F^\zeta)t} \|v_F^\varepsilon(t) - h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_S^\varepsilon(t))\|_{Y_1} \\
& \quad + C_A L_f (\varepsilon L_g + 1) \int_0^t \frac{e^{\varepsilon^{-1}(\varepsilon\omega_A - \lambda_0)(t-s)}}{\varepsilon^\gamma (t-s)^{1-\gamma}} z(s) ds \\
& \quad + C_B L_g (L_g + 1) \int_0^t \frac{e^{(\zeta^{-1}(\varepsilon\omega_A + \lambda_0) + N_F^\zeta)(t-s)}}{(t-s)^{1-\delta}} z(s) ds.
\end{aligned}$$

Applying a version of Gronwall's inequality yields

$$z(t) \leq C e^{-ct} \left\| \begin{pmatrix} u_0 - h_{X_1}^{\varepsilon, \zeta}(v_{0,S}) \\ v_{0,F} - h_{X_1}^{\varepsilon, \zeta}(v_{0,S}) \end{pmatrix} \right\|_{X_1 \times Y_1}.$$

This completes the proof. $\square$

The last step in the generalization of Fenichel's theorem is to show that the semi-slow flow is a good approximation of the semi-flow of the original fast-reaction system.

So far we have measured the distance of the slow manifold to the subset $S_{0,\zeta}$ of the critical manifold given by

$$S_{0,\zeta} := \{(h^0(v_0), v_0) \in S_0 : \operatorname{pr}_{Y_F^\zeta} v_0 = 0\}.$$

However, this subset is in general not invariant under the slow flow. Hence, we introduce the reduced slow subsystem

$$\begin{aligned}
0 & = f(h^0(v_\zeta^0(t)), v_\zeta^0(t)), \\
0 & = \operatorname{pr}_{Y_F^\zeta} v_\zeta^0(t), \\
\partial_t v_\zeta^0(t) & = B v_\zeta^0(t) + \operatorname{pr}_{Y_S^\zeta} g(h^0(v_\zeta^0(t)), v_\zeta^0(t)), \\
v_\zeta^0(0) & = \operatorname{pr}_{Y_S^\zeta} v_0,
\end{aligned} \tag{3.10}$$

and we observe that $S_{0,\zeta}$ is forward invariant under the flow generated by the slow subsystem.

Proposition 3.10. *For all $T > 0$, there exists a constant $C > 0$ such that for all $t \in [0, T]$ and all $\zeta > 0$ small enough the following holds*

$$\|v^0(t) - v_\zeta^0(t)\|_{Y_1} \leq C \left(\|\operatorname{pr}_{Y_F^\zeta} v_0\|_{Y_1} + \frac{\|v_0\|_{Y_1}}{(\omega_B - \zeta^{-1}\lambda_0 - N_F^\zeta)^\delta} \right).$$

Proof. The well-posedness of the slow subsystem follows from Prop 2.6 and thus we can write the solution as

$$v_\zeta^0(t) = e^{tB} \text{pr}_{Y_S^\zeta} v_0 + \int_0^t e^{(t-s)B} \text{pr}_{Y_S^\zeta} g(h^0(v_\zeta^0(s)), v_\zeta^0(s)) \, ds.$$

Hence

$$\begin{aligned} v^0(t) - v_\zeta^0(t) &= \\ &= e^{tB} \text{pr}_{Y_F^\zeta} v_0 + \int_0^t e^{(t-s)B} \text{pr}_{Y_S^\zeta} (g(h^0(v^0(s)), v^0(s)) - g(h^0(v_\zeta^0(s)), v_\zeta^0(s))) \, ds \\ &\quad - \int_0^t e^{(t-s)B} \text{pr}_{Y_F^\zeta} g(h^0(v_\zeta^0(s)), v_\zeta^0(s)) \, ds. \end{aligned}$$

Now, we can estimate

$$\begin{aligned} \|v^0(t) - v_\zeta^0(t)\|_{Y_1} &\leq \\ &\leq M_B e^{(\zeta^{-1}\lambda_0 + N_F^\zeta)t} \|\text{pr}_{Y_F^\zeta} v_0\|_{Y_1} \\ &\quad + L_g C_B \int_0^t \frac{e^{(\zeta^{-1}\lambda_0 + N_F^\zeta)(t-s)}}{(t-s)^{1-\delta}} (\|h^0(v^0(s))\|_{X_1} + \|v^0(s)\|_{Y_1}) \, ds \\ &\quad + L_G C_B \int_0^t \frac{e^{\omega_B(t-s)}}{(t-s)^{1-\delta}} (\|h^0(v^0(s)) - h^0(v_\zeta^0(s))\|_{X_1} + \|v^0(s) - v_\zeta^0(s)\|_{Y_1}) \, ds \\ &\leq M_B e^{(\zeta^{-1}\lambda_0 + N_F^\zeta)t} \|\text{pr}_{Y_F^\zeta} v_0\|_{Y_1} \\ &\quad + L_g (L_{h^0} + 1) M_B C_B e^{C_B L_g (1+L_{h^0})\Gamma(\delta)} e^{\omega_B t} \|v_0\|_{Y_1} \int_0^t \frac{e^{(\zeta^{-1}\lambda_0 + N_F^\zeta - \omega_B)(t-s)}}{(t-s)^{1-\delta}} \, ds \\ &\quad + L_G (L_{h^0} + 1) C_B \int_0^t \frac{e^{\omega_B(t-s)}}{(t-s)^{1-\delta}} \|v^0(s) - v_\zeta^0(s)\|_{Y_1} \, ds \\ &\leq C e^{\omega_B t} \left(\|\text{pr}_{Y_F^\zeta} v_0\|_{Y_1} + \frac{\|v_0\|_{Y_1}}{(\omega_B - \zeta^{-1}\lambda_0 - N_F^\zeta)^\delta} \right) \\ &\quad + L_G (L_{h^0} + 1) C_B \int_0^t \frac{e^{\omega_B(t-s)}}{(t-s)^{1-\delta}} \|v^0(s) - v_\zeta^0(s)\|_{Y_1} \, ds. \end{aligned}$$

Applying Gronwall's inequality yields the desired result. $\square$

Remark 3.11. In the case that the operator B is dissipative, i.e. $\|e^{tB}\|_{\mathcal{B}(Y_1)} \leq e^{\omega_B t}$ with $\omega_B < 0$ the above result reduces to

$$\|v^0(t) - v_\zeta^0(t)\|_{Y_1} \leq C e^{\omega_B t} \left(\|\text{pr}_{Y_F^\zeta} v_0\|_{Y_1} + \frac{\|v_0\|_{Y_1}}{|\omega_B - \zeta^{-1}\lambda_0 - N_F^\zeta|^\delta} \right).$$

Proposition 3.12. For all $T > 0$ there exists a constant $C > 0$ such that for all $t \in [0, T]$ and all $\zeta > 0$ small enough it holds that

$$\begin{aligned} &\left\| \begin{pmatrix} u^\varepsilon(t) - h^0(v_\zeta^0(t)) \\ v^\varepsilon(t) - v_\zeta^0(t) \end{pmatrix} \right\|_{X_\alpha \times Y_1} \leq \\ &\leq C e^{\omega_B t + ct\delta} \varepsilon^{1-\alpha} (e^{\omega_B t} + \varepsilon^{-\gamma} t^\gamma e^{\varepsilon^{-1}(\varepsilon\omega_A + \lambda_0)t}) \|v_0\|_{Y_1} \end{aligned}$$

$$\begin{aligned}
& + C e^{\varepsilon \omega_B t + c t} C M_A e^{\varepsilon^{-1}(\varepsilon \omega_A + \lambda_0)t} \|u_0 - h^0(v_0)\|_{X_1} \\
& + C e^{\omega_B t} e^{\varepsilon \omega_B t} \left(\|pr_{Y_F^\zeta} v_0\|_{Y_1} + \frac{\|v_0\|_{Y_1}}{(\omega_B - \zeta^{-1} \lambda_0 - N_F^\zeta)^\delta} \right).
\end{aligned}$$

If in addition the initial values are on the slow manifold it holds that

$$\begin{aligned}
& \left\| \begin{pmatrix} u^\varepsilon(t) - h^0(v_\zeta^0(t)) \\ v^\varepsilon(t) - v_\zeta^0(t) \end{pmatrix} \right\|_{X_\alpha \times Y_1} \leq C e^{(\omega_B + c)t} \|v_0\|_{Y_1} \\
& \times \left(\varepsilon^{1-\alpha} t^\delta + \varepsilon^{-\gamma} t^\gamma e^{\varepsilon^{-1}(\varepsilon \omega_A + \lambda_0)t} + \frac{1}{(\omega_B - \zeta^{-1} \lambda_0 - N_F^\zeta)^\delta} + \frac{1}{(N_S^\zeta - N_F^\zeta)^\delta} \right).
\end{aligned}$$

Proof. The first estimate is a consequence of Theorem 2.14 and Proposition 3.9. For the second estimate we apply the triangle inequality to the terms

$$\begin{aligned}
\|pr_{Y_F^\zeta} v_0\|_{Y_1} & \leq \|pr_{Y_F^\zeta} v_0 - h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0)\|_{Y_1} + \|h_{Y_F^\zeta}^{\varepsilon, \zeta}(v_0)\|_{Y_1}, \\
\|u_0 - h^0(v_0)\|_{X_1} & \leq \|u_0 - h_{X_1}^{\varepsilon, \zeta}(v_0)\|_{X_1} + \|h_{X_1}^{\varepsilon, \zeta}(v_0) - h^0(v_0)\|_{X_1}.
\end{aligned}$$

The first term in each of the two estimates equals to zero since the initial data is on the slow manifold and the second term can be estimated by Proposition 3.6. This then proofs the fourth and last statement (GF4) of the generalized Fenichel Theorem 3.1. $\square$

Remark 3.13. Again we want to put the final result into context. To this end we assume that the constant N_F^ζ is bounded independent of ζ and that in the asymptotic limit $N_S^\zeta - N_F^\zeta = C\zeta^{-\beta}$. Then, using the assumption that $\varepsilon\zeta^{-1} < 1$ we obtain

$$\begin{aligned}
& \left\| \begin{pmatrix} u^\varepsilon(t) - h^0(v_\zeta^0(t)) \\ v^\varepsilon(t) - v_\zeta^0(t) \end{pmatrix} \right\|_{X_\alpha \times Y_1} \leq C e^{(\omega_B + c)t} \|v_0\|_{Y_1} \\
& \times (\varepsilon^{\alpha-1} + \varepsilon^{-\gamma} e^{\varepsilon^{-1}(\varepsilon \omega_A + \lambda_0)t} + \zeta^\beta + \zeta^{\beta\delta}).
\end{aligned}$$

Letting $\varepsilon, \zeta \rightarrow 0$ yields an explicit rate for the convergence of the semi-flow fast-reaction system to the semi-flow of the slow subsystem.

3.5. Example

As a final step we apply our method to a model from biology, the Beddington-DeAngelis type predator-prey model [4, 10]. Let $\Omega \subset \mathbb{R}^n$, then the system reads

$$\begin{aligned}
\partial_t p_h^\varepsilon &= d_h \Delta p_h^\varepsilon + \frac{1}{\varepsilon} (aN^\varepsilon p_s^\varepsilon - \tilde{\gamma} p_h^\varepsilon) - \mu p_h^\varepsilon, \\
\partial_t p_s^\varepsilon &= d_s \Delta p_s^\varepsilon - \frac{1}{\varepsilon} (aN^\varepsilon p_s^\varepsilon - \tilde{\gamma} p_h^\varepsilon) - \mu p_s^\varepsilon + \Gamma p_h^\varepsilon, \\
\partial_t N^\varepsilon &= d_N \Delta N^\varepsilon + r_0(1 - \eta N^\varepsilon)N^\varepsilon - aN^\varepsilon p_s^\varepsilon,
\end{aligned}$$

subject to homogeneous Neumann boundary conditions.

To bring this system in the form of (2.3) we introduce a new variable $P^\varepsilon = p_h^\varepsilon + p_s^\varepsilon$ and add the first and second equation to obtain the evolution of

P^ε . Moreover, we assume that the diffusion coefficients satisfy $d_h = d_s$. This ensures the semi-linear structure of the equations and we have

$$\begin{aligned}\partial_t p_h^\varepsilon &= d\Delta p_h^\varepsilon + \frac{1}{\varepsilon} \left(aN^\varepsilon(P^\varepsilon - p_h^\varepsilon) - \tilde{\gamma}p_h^\varepsilon \right) - \mu p_h^\varepsilon, \\ \partial_t P^\varepsilon &= d\Delta P^\varepsilon - \mu P^\varepsilon + \Gamma p_h^\varepsilon, \\ \partial_t N^\varepsilon &= d_N \Delta N^\varepsilon + r_0(1 - \eta N^\varepsilon)N^\varepsilon - aN^\varepsilon(P^\varepsilon - p_h^\varepsilon).\end{aligned}\tag{3.11}$$

The limiting system as $\varepsilon \rightarrow 0$ has the form

$$\begin{aligned}0 &= aN^0(P^0 - p_h^0) - \tilde{\gamma}p_h^0, \\ \partial_t P^0 &= d\Delta P^0 - \mu P^0 + \Gamma p_h^0, \\ \partial_t N^0 &= d_N \Delta N^0 + r_0(1 - \eta N^0)N^0 - aN^0(P^0 - p_h^0).\end{aligned}\tag{3.12}$$

The case where $d_h \neq d_s$ is studied in an upcoming work. We note that in [10, 52] the convergence of solutions of (3.11) to (3.12) in some L^p -spaces was shown using entropy methods. However, we seek the convergence in more regular spaces and also aim to construct the slow manifold in these more regular spaces.

Moreover, as we require global Lipschitz continuity in the nonlinear terms we modify the above system. Let $\nu > 0$ be a small parameter. Then, the modified system reads

$$\begin{aligned}\partial_t p_h^\varepsilon &= (d\Delta - \mu \text{Id})p_h^\varepsilon + \frac{1}{\varepsilon} \left(a \frac{N^\varepsilon(P^\varepsilon - p_h^\varepsilon)}{1 + \nu((N^\varepsilon)^2 + (P^\varepsilon)^2)} - \tilde{\gamma}p_h^\varepsilon \right), \\ \partial_t P^\varepsilon &= d\Delta P^\varepsilon - \mu P^\varepsilon + \Gamma p_h^\varepsilon, \\ \partial_t N^\varepsilon &= d_N \Delta N^\varepsilon + r_0(1 - \eta N^\varepsilon)N^\varepsilon - a \frac{N^\varepsilon(P^\varepsilon - p_h^\varepsilon)}{1 + \nu((N^\varepsilon)^2 + (P^\varepsilon)^2)}.\end{aligned}\tag{3.13}$$

Here, the new limiting system as $\varepsilon \rightarrow 0$ has the form

$$\begin{aligned}0 &= a \frac{N^0(P^0 - p_h^0)}{1 + \nu((N^0)^2 + (P^0)^2)} - \tilde{\gamma}p_h^0, \\ \partial_t P^0 &= d\Delta P^0 - \mu P^0 + \Gamma p_h^0, \\ \partial_t N^0 &= d_N \Delta N^0 + r_0(1 - \eta N^0)N^0 - a \frac{N^0(P^0 - p_h^0)}{1 + \nu((N^0)^2 + (P^0)^2)}.\end{aligned}\tag{3.14}$$

Now, let $\Omega = \mathbb{T}^n = [0, 2\pi]^n$ with $n = 1, 2, 3$ and let $X = Y = L^2(\Omega)$ be the underlying Banach space with $X_1 = Y_1 = H^2(\Omega)$. Moreover, let the initial data $N_0, P_0, p_{h,0} \in H^2(\Omega)$ satisfy the critical manifold constraint

$$-\tilde{\gamma}p_{h,0} + a \frac{N_0(P_0 - p_{h,0})}{1 + \nu((N_0)^2 + (P_0)^2)} = 0.\tag{3.15}$$

Theorem 3.14. *Let $(p_h^\varepsilon, P^\varepsilon, N^\varepsilon)$ be the solution of (3.11) and let (p_h^0, P^0, N^0) be the solution of the limit system (3.12). Then there exist constants $C, c > 0$ such that for all $\varepsilon \in (0, \varepsilon_0]$ and all $N_0, P_0, p_{h,0} \in H^2(\Omega)$ satisfying (3.15) with $\|N_0\|_{H^2(\Omega)}, \|P_0\|_{H^2(\Omega)}, \|p_{h,0}\|_{H^2(\Omega)} \leq \sigma$ it holds*

$$\sup_{0 \leq t \leq T(\sigma)} \left(\|p_h^\varepsilon(t) - p_h^0(t)\|_{H^{2\alpha}(\Omega)} + \|P^\varepsilon(t) - P^0(t)\|_{H^2(\Omega)} + \|N^\varepsilon(t) - N^0(t)\|_{H^2(\Omega)} \right) \leq C(\varepsilon^{1-\alpha} + e^{-\varepsilon^{-1}ct}),$$

where $\alpha \in [0, 1)$ such that $2\alpha > \frac{n}{2}$ and the final time $T(\sigma)$ is given by

$$T(\sigma) := \inf \{t \in [0, \infty) : \max\{\|p_h^\varepsilon(t)\|_{H^2(\Omega)}, \|P^\varepsilon(t)\|_{H^2(\Omega)}, \|N^\varepsilon(t)\|_{H^2(\Omega)}, \|p_h^0(t)\|_{H^2(\Omega)}, \|P^0(t)\|_{H^2(\Omega)}, \|N^0(t)\|_{H^2(\Omega)}\} > \sigma\}.$$

Theorem 3.15. *Let $n = 1$ and consider the domain $\Omega = \mathbb{T} = [0, 2\pi]$ and let the assumptions of the previous result hold. Moreover, there is a ζ_0 such that for all $\zeta \in (0, \zeta_0]$ satisfying $\varepsilon\zeta^{-1} < \frac{1}{2}$ the following holds:*

- *There exists a slow manifold $S_{\varepsilon, \zeta}$ that is C^1 -regular and locally invariant under the semi-flow generated by (3.11).*
- *Let $S_{0, \zeta} := \{(p_h, P, N) \in S_0 : (P, N) \in B_{Y_S^\zeta}(0, \sigma)\}$ be the submanifold of the critical manifold which consists of all points whose slow components are in the bounded set $B_{Y_S^\zeta}(0, \sigma)$. Then there exists a constant $C > 0$ such that*

$$\text{dist}(S_{\varepsilon, \zeta}, S_{0, \zeta}) \leq C(\varepsilon^{1-\alpha} + \zeta^{1/2}).$$

- *Suppose that $(h^0(P_0, N_0), P_0, N_0) \in S_{\varepsilon, \zeta}$ satisfying*

$$\|P_0\|_{H^2([0, 2\pi])}, \|N_0\|_{H^2([0, 2\pi])}, \|h^0(P_0, N_0)\|_{H^2([0, 2\pi])} \leq \sigma.$$

Then there exists a constant $C > 0$ such that

$$\sup_{0 \leq t \leq T(\sigma)} \left(\|p_h^\varepsilon(t) - p_h^0(t)\|_{H^{2\alpha}(\Omega)} + \|P^\varepsilon(t) - P^0(t)\|_{H^2(\Omega)} + \|N^\varepsilon(t) - N^0(t)\|_{H^2(\Omega)} \right) \leq C(\zeta^{1-\alpha} + \zeta^{1/2}).$$

What remains is to check the assumptions from sections 2.1 and 3.1.

- The Laplacian generates an analytic C_0 -semigroup $(e^{t\Delta})_{t \geq 0}$ on $X_1 = H^2(\Omega) \subset L^2(\Omega)$ and from complex interpolation it follows that the spaces $X_\alpha = H^{2\alpha}(\Omega)$ are a suitable choice for the Banach scales. By restricting $\alpha \in (\frac{n}{4}, 1)$ it follows that $H^{2\alpha}(\Omega)$ and $H^2(\Omega)$ are Banach algebras.
- In this example we have $A = B_1 = d\Delta - \mu\text{Id}$, which implies that the growth bound satisfies $\omega_A = \omega_{B_1} \leq -\mu$. Similarly, we have $B_2 = d_N\Delta$ and $\omega_{B_2} \leq 0$.
- The nonlinearities of the system are

$$f(p_h, P, N) = a \frac{N(P - p_h)}{1 + \nu(N^2 + P^2)} - \tilde{\gamma}p_h,$$

$$g_1(p_h, P, N) = \Gamma p_h, \quad \text{and}$$

$$g_2(p_h, P, N) = r_0(1 - \eta N)N - a \frac{N(P - p_h)}{1 + \nu(N^2 + P^2)}.$$

As the nonlinearities do not decrease the regularity we set $\gamma = \delta = 1$. In addition, the function f is Fréchet differentiable as a mapping $f : H^{2\alpha} \times H^{2\alpha} \times H^{2\alpha} \rightarrow H^{2\alpha}$ and similar the functions f, g_1, g_2 are Fréchet differentiable as mappings $f, g_1, g_2 : H^2 \times H^2 \times H^2 \rightarrow H^2$. As $0 \leq p_h \leq P$ by the definition of P it follows that there exists a constant $L_f > 0$ such that

$$\begin{aligned} \|Df(x, y, z)\|_{\mathcal{B}(H^2 \times H^2 \times H^2, H^2)} &\leq L_f, \\ \|Df(x, y, z)\|_{\mathcal{B}(H^{2\alpha} \times H^{2\alpha} \times H^{2\alpha}, H^{2\alpha})} &\leq L_f. \end{aligned}$$

This implies that the following inequalities hold globally

$$\begin{aligned} \|f(x_1, y_1, z_1) - f(x_2, y_2, z_1)\|_{H^2} &\leq \\ &\leq L_f (\|x_1 - x_2\|_{H^2} + \|y_1 - y_2\|_{H^2} + \|z_1 - z_2\|_{H^2}), \\ \|f(\tilde{x}_1, \tilde{y}_1, \tilde{z}_1) - f(\tilde{x}_2, \tilde{y}_2, \tilde{z}_1)\|_{H^{2\alpha}} &\leq \\ &\leq L_f (\|\tilde{x}_1 - \tilde{x}_2\|_{H^{2\alpha}} + \|\tilde{y}_1 - \tilde{y}_2\|_{H^{2\alpha}} + \|\tilde{z}_1 - \tilde{z}_2\|_{H^{2\alpha}}), \end{aligned}$$

where $x_1, x_2, y_1, y_2, z_1, z_2 \in H^2(\Omega)$ and $\tilde{x}_1, \tilde{x}_2, \tilde{y}_1, \tilde{y}_2, \tilde{z}_1, \tilde{z}_2 \in H^{2\alpha}$ and where the Lipschitz constant is bounded by $L_f \leq \max\{a\nu^{-1}, \tilde{\gamma}\}$.

Although in [10] and [52] it was shown that N has a global L^∞ -bound N_∞ , this is not enough to guarantee the global Lipschitz continuity for g_2 in the modified systems (3.13). To ensure this we apply a cutoff technique. Let $\sigma > 0$ and choose a C^∞ -cutoff function $\chi : H^2(\Omega) \rightarrow [0, 1]$ such that $\chi(x) = 1$ if $x \in B(0, \sigma)$, $\chi(x) = 0$ if $x \in H^2(\Omega) \setminus B(0, 2\sigma)$ and $\|D\chi\|_{\mathcal{B}(H^2(\Omega), \mathbb{R})} \leq \sigma$. Then we have that

$$\begin{aligned} g_{2,\chi} : H^2 \times H^2 \times H^2 &\rightarrow H^2, \\ (x, y, z) &\mapsto r_0(1 - \eta\chi(\|z\|_{H^2})z)\chi(\|z\|_{H^2})z - a \frac{z(y-x)}{1 + \nu(z^2 + y^2)} \end{aligned}$$

satisfies

$$\|Dg_{2,\chi}(x, y, z)\|_{\mathcal{B}(H^2 \times H^2 \times H^2, H^2)} \leq L_{g_2}.$$

which implies

$$\begin{aligned} \|g_{2,\chi}(x_1, y_1, z_1) - g_{2,\chi}(x_2, y_2, z_1)\|_{H^2} &\leq \\ &\leq L_{g_2} (\|x_1 - x_2\|_{H^2} + \|y_1 - y_2\|_{H^2} + \|z_1 - z_2\|_{H^2}) \end{aligned}$$

for $x_1, x_2, y_1, y_2, z_1, z_2 \in H^2(\Omega)$.

- (iv) We compute that $D_{p_h}f = -aN - \tilde{\gamma} \leq -\tilde{\gamma} < 0$ as solutions to (3.14) satisfy $N \geq 0$. Hence, the nonlinearity f satisfies the assumptions of the implicit function theorem and we obtain the existence of a function $h^0 : H^2 \times H^2 \rightarrow H^2$ such that $h^0(P^0, N^0) = p_h^0$ for all $(P^0, N^0) \in V \subset H^2 \times H^2$.
- (v) The function h^0 given by the implicit function theorem can be computed explicitly, i.e.

$$h^0(P^0, N^0) = \frac{aN^0P^0}{\tilde{\gamma} + aN^0 + \tilde{\gamma}\nu((N^0)^2 + (P^0)^2)}.$$

From this explicit form we immediately see that $h^0 : H^2 \times H^2 \rightarrow H^2$ is Lipschitz continuous with constant L_{h^0} .

Thus the limit system (3.14) can be written as

$$\begin{aligned}\partial_t P^0 &= d\Delta P^0 - \mu P^0 + \Gamma \frac{aN^0 P^0}{\tilde{\gamma} + aN^0 + \tilde{\gamma}\nu((N^0)^2 + (P^0)^2)}, \\ \partial_t N^0 &= d_N \Delta N^0 + r_0(1 - \eta N^0)N^0 - \frac{aN \left(P - \frac{aN^0 P^0}{\tilde{\gamma} + aN^0 + \tilde{\gamma}\nu((N^0)^2 + (P^0)^2)} \right)}{1 + \nu((N^0)^2 + (P^0)^2)},\end{aligned}\quad (3.16)$$

defined on the slow manifold

$$S_0 = \left\{ (p_h, P, N) \in (H^2)^3 : p_h = \frac{aN^0 P^0}{\tilde{\gamma} + aN^0 + \tilde{\gamma}\nu((N^0)^2 + (P^0)^2)} \right\}. \quad (3.17)$$

(vi) Using that $D_{p_h} f(p_h, P, N) \leq -\tilde{\gamma}$, we can introduce the operator $\tilde{A}_\varepsilon$ as

$$\tilde{A}_\varepsilon = \varepsilon d\Delta - (\varepsilon\mu + \tilde{\gamma})\text{Id}$$

and the corresponding nonlinear function

$$\tilde{f} = a \frac{N(P - p_h)}{1 + \nu(N^2 + P^2)}.$$

It follows that $\tilde{A}_\varepsilon$ generates an analytic semigroup with growth bound $\varepsilon(\omega_A - \mu) - \tilde{\gamma} < 0$. Moreover, we have that the Lipschitz constant of $\tilde{f}$ is bounded by $L_{\tilde{f}} \leq a\nu^{-1}$.

Thus, all assumptions in section 2.1 are satisfied and it remains to show that the Laplacian satisfies the assumptions for the splitting.

Making use of the Hilbert structure of $L^2([0, 2\pi])$ we introduce the splitting as

$$L^2 = Y = Y_S^\zeta \oplus Y_F^\zeta$$

by truncating at a certain Fourier mode. Assuming that

$$-(k_0 + 1)^2 < \zeta^{-1}(\varepsilon(\omega_A - \mu) - \tilde{\gamma}) \leq -k_0^2$$

we set

$$Y_S^\zeta = \text{span}\{x \mapsto e^{ikx} : |k| \leq k_0\}, \quad Y_F^\zeta = \text{cl}_{L^2}(\text{span}\{x \mapsto e^{ikx} : |k| > k_0\}).$$

(vii) Since Y_S^ζ is a finite-dimensional space the Laplacian $\Delta - \mu\text{Id}$ generates a C_0 -group on Y_S^ζ , where $(e^{t(\Delta - \mu\text{Id})}|_{Y_S^\zeta})_{t \in \mathbb{R}}$ coincides with $(e^{t(\Delta - \mu\text{Id})})|_{Y_S^\zeta}$ for $t \geq 0$.

(viii) For the realization of the Laplacian in Y_F^ζ we have that 0 is in its resolvent. This can be seen by the representation of the inverse, which is given by

$$\Delta_{Y_F^\zeta}^{-1} : L^2([0, 2\pi]) \rightarrow H^2([0, 2\pi]), \quad \sum_{|k| \geq k_0} \hat{f}(k) e^{ikx} \mapsto \sum_{|k| \geq k_0} \frac{\hat{f}(k) e^{ikx}}{|k|^2}$$

and is well-defined as $k = 0$ does not appear in the sum.

(ix) We note that the semigroup $(e^{t\Delta})_{t \geq 0}$ can be expressed as

$$e^{t(\Delta - \mu \text{Id})} f = \left[x \mapsto \sum_{k \in \mathbb{Z}} e^{-(|k|^2 + \mu)t} \hat{f}(k) e^{ikx} \right].$$

Applying Plancherel's theorem for $y_S \in Y_S^\zeta$ and $t \geq 0$ yields

$$\|e^{-t(\Delta - \mu \text{Id})} y_S\|_{H^2([0, 2\pi])} \leq e^{(k_0 - 1)^2 t} \|y_S\|_{H^2([0, 2\pi])}.$$

Thus, we can take $N_S^\zeta = \zeta^{-1} \tilde{\gamma} - (k_0 - 1)^2$ and similarly we may take $N_F^\zeta = \zeta^{-1} \tilde{\gamma} - k_0^2$. Then,

$$N_S^\zeta - N_F^\zeta = 2k_0 - 1 \geq 2\sqrt{\zeta^{-1} \tilde{\gamma}} - C.$$

Therefore, $N_S^\zeta - N_F^\zeta \approx \mathcal{O}(\zeta^{-1/2})$ for ζ small enough.

(x) The last assumption we need to check is the spectral gap condition (3.1). To this end let $\varepsilon \zeta^{-1} \leq \frac{1}{2}$ and assume that the parameter ν is chosen large enough such that $C_A a \tilde{\gamma} < \frac{1}{2} \nu$. Then, $C(L_{g_1} + L_{g_2}) < \frac{1}{2}(N_S^\zeta - N_F^\zeta)$ holds for all sufficiently small $\zeta > 0$ and the spectral gap condition is satisfied.

This completes checking the assumptions for section 3.1.

Remark 3.16. The dissipative structure of the system (3.11) is key that the we do not loose to much information on the solutions when we apply the cut-off method to obtain global Lipschitz bounds.

Remark 3.17. We want to emphasize that the slow manifold for the original system exist only in the region of the phase space that satisfies the uniform bound already prior to the modification and cut-off procedure. This means that in the application at hand, the exponential attraction towards the slow manifold (cf. Proposition 3.9) is only valid in the aforementioned region of phase space. Hence, to see the slow manifold in numerical simulations it is crucial for the initial data to be already close to the critical manifold.

4. Discussion and outlook

We conclude this paper with a discussion of the results and an outlook to open problems we believe to be important to investigate.

The theory of slow manifolds for multiscale PDEs is closely linked to other areas of invariant manifold theory for PDEs, e.g., to center manifolds [28, 55] and inertial manifolds [14, 54, 57]. In principle, the mathematical methods to show the existence of invariant manifolds are similar for all approaches. The classical choices are Lyapunov-Perron, graph transform, or implicit function theorem methods to find an invariant manifold. Yet, there are a couple of crucial differences in the setup for geometric singular perturbation theory. First, one wants to have localized manifolds that can be patched as one usually also requires to track changes between the fast and slow subsystems. Second, one seeks an understanding of the convergence of the slow manifold $S_{\varepsilon, \zeta}$ to the critical manifold S_0 in both a geometric and analytical way.

Now, we want to explain the above mentioned convergence in more detail. The critical manifold $S_0 \subset X \times Y$ given by $S_0 = \{(u, v) \in X \times Y : f(u, v) = 0\}$ is a (possibly) infinite dimensional manifold in the phase space on which the slow dynamics evolve. With the splitting of Y into a fast component Y_F^ζ and a slow component Y_S^ζ we obtain that the slow component has a finite dimension, where $\dim Y_S^\zeta$ is related to ζ^{-1} . As the slow manifold is constructed as a function $(h_{X_1}^{\varepsilon, \zeta}(v_S), h_{X_F}^{\varepsilon, \zeta}(v_S))$ over the slow variable space it follows that $\dim S_{\varepsilon, \zeta} \leq 2 \dim Y_S^\zeta$. Thus, with decreasing ζ the dimension of the slow variable space and therefore also of the slow manifold increases and in the limit $\zeta \rightarrow 0$ coincides with the critical manifold.

In applications one might need to construct the explicit slow manifold. However, as the current existence proof of the slow manifold is non-constructive the question is whether this is possible. So far, we have only shown a construction of the slow manifold for linear fast-slow systems [44]. One method to obtain an approximate slow manifolds is via a Galerkin approximation and the ideas presented in [17] for fast-slow systems also hold in the case of fast-reaction equations.

Another open question concerns the spectral gap condition. As in the theory for inertial manifolds, the question whether there are slow manifolds in the case when the spectral gap condition is violated remains open. We conjecture that the spectral gap condition is only a sufficient condition for the existence of the slow manifold and that there are cases similar to the one in [57] that give us examples of existence but also counterexamples.

We note some differences between the results obtained for fast-reaction systems in this work and the results in [35] for fast-slow systems. One key issue are the restrictions on the operator A . In this work the growth bound on the semigroup generated by the operator A need not satisfy $\omega_A < 0$ since the fast-reaction dominates the terms on the right-hand side. Therefore, we only require that $\|e^{\varepsilon^{-1}(\varepsilon A - D_u f(h^0(v^0), v^0))t}\|_{B(X_1)} \leq M_A e^{\varepsilon^{-1}(\varepsilon \omega_A - \lambda_0)t}$ where $\varepsilon \omega_A - \lambda_0 < 0$. On the other hand, we need more requirements on the nonlinear function f , one of which is, that f has to satisfy the implicit function theorem in a neighborhood around the equilibrium $f(u, v) = 0$. Another difference is that, with the current method, we loose regularity in the estimates for the slow manifold, i.e. the distance result (Prop. 3.6) and the convergence results (Prop. 3.9) only hold in the space $X_\alpha \times Y_1$ and not in $X_1 \times Y_1$.

So far, the structure of the fast-slow PDE systems considered in this and the previous results [35, 44] is that the functions f, g are independent of the parameter ε . However, we believe that it is of interest to also consider fast-slow systems beyond the standard form, see [56] for an introduction in the finite dimensional case or [39] for a concrete ODE example.

From the viewpoint of fast-slow PDEs, it would be natural to also study fast reaction systems with a loss of normal hyperbolicity, i.e., to allow for more complex bifurcation scenarios between fast and slow variables. For fast-slow PDEs in standard form, this theory is beginning to emerge. Examples on a bounded domain are the transcritical case in [20] and the fold case in [18]. For

unbounded domains, there is a recent result on the transcritical and pitchfork case [38]. For unbounded domains, there is a natural connections to fast-slow PDEs and their reduced dynamic amplitude equations [34, 37], where the lack of a spectral gap is compensated via a suitable localization of the critical modes.

Lastly, we mention that one should also consider the further technical development of the proof strategy. For example, in Appendix B we present an alternative construction of the slow manifold that could be of independent interest.

Acknowledgements

The authors would like to thank F. Hummel, B.Q. Tang and B.-N. Tran for the fruitful discussions. We also thank the referees for several comments which improved the presentation of the results. Furthermore, the authors acknowledge the support of the DFG under grant No.456754695.

Funding Open Access funding enabled and organized by Projekt DEAL.

Open Access. This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Appendix A. Tools from functional analysis

In this section we present useful definitions and theorems from the theory of semigroups that are used throughout the paper. The details to the brief summary we present here can be found in [1] and [45] and the references therein.

Definition A.1. Let X be a Banach space. Then a family of linear, bounded operators $(S(t))_{t \geq 0} \subset \mathcal{L}(X)$, where $S(t) : X \rightarrow X$ for all $t \geq 0$, is called a semigroup if

$$S(0) = I, \quad S(s+t) = S(s)S(t) \quad \forall t \geq 0.$$

It is called a strongly continuous or C_0 -semigroup if in addition

$$\|S(t)x - x\|_X \rightarrow 0, \quad t \rightarrow 0^+ \quad \forall x \in X.$$

Definition A.2. Define the set $D(A) := \{x \in X : \exists \lim_{t \rightarrow 0+} \frac{S(t)x - x}{t}\}$ and $Ax := \lim_{t \rightarrow 0+} \frac{S(t)x - x}{t}$ for $x \in D(A)$. Then, A is called the infinitesimal generator of $(S(t))_{t \geq 0}$.

Theorem A.3. Let $x \in D(A)$ and assume that $\sup_{t \geq 0} \|S(t)\| < \infty$. Then

- i) $S(t)x \in D(A)$ for all $t \geq 0$.
- ii) $AS(t)x = S(t)Ax$ for all $t \geq 0$.
- iii) The mapping $t \mapsto S(t)x$ is differentiable for each $t \geq 0$ and $\frac{d}{dt}S(t)x = AS(t)x$ for all $t \geq 0$.

Theorem A.4. Let $(S(t))_{t \geq 0}$ be a C_0 -semigroup on X . Then there exists an $\omega \in \mathbb{R}$ and $M \geq 1$ such that

$$\|S(t)\|_{\mathcal{B}(X)} \leq Me^{\omega t} \quad \forall t \geq 0.$$

Theorem A.5. Let A be the generator of a C_0 -semigroup $(S(t))_{t \geq 0}$ on X . If $B \in \mathcal{B}(X)$, then $A+B$ is the infinitesimal generator of a C_0 -semigroup $(T(t))_{t \geq 0}$ on X satisfying

$$\|T(t)\|_{\mathcal{B}(X)} \leq Me^{(\omega+M\|B\|_{\mathcal{B}(X)})t} \quad \forall t \geq 0.$$

Theorem A.6. Let A be the generator of a C_0 -semigroup $(S(t))_{t \geq 0}$ on X and let $x \in D(A)$. Then the mapping $t \mapsto S(t)x$ solves the abstract Cauchy problem on X for $t \geq 0$

$$\frac{d}{dt}y = Ay, \quad y(0) = x.$$

Definition A.7. Let A be a densely defined closed linear operator on X with $0 \in \varrho(A)$. Let $\theta \in (0, 1)$. Then we call $(\cdot, \cdot)_\theta$ an exact admissible interpolation functor, i.e. it is an exact interpolation functor such that X_1 is dense in $(X_0, X_1)_\theta$ whenever $X_1 \xhookrightarrow{d} X_0$.

Definition A.8. We define a family of Banach spaces $(X_\alpha)_{\alpha \in [-1, \infty)}$ and a family of operators $(A^\alpha)_{\alpha \in [-1, \infty)} \in \mathcal{B}(X_{\alpha+1}, X_\alpha)$ as follows

- for $k \in \mathbb{N}_0$ we set $X_k := D(A^k)$ endowed with the norm $\|x\|_{X_k} := \|A^k x\|_X$. In particular, $X_0 = D(A^0) = X$;
- X_{-1} is defined as the completion of X with respect to the norm $\|x\|_{X_{-1}} := \|A^{-1}x\|_X$;
- For $k \in \mathbb{N}_0$, $\theta \in (0, 1)$ and $\alpha = k + \theta$ we define $X_\alpha := (X_k, X_{k+1})_\theta$ and $A^\alpha = A^k|_{D(A^\alpha)}$, where $D(A^\alpha) = \{x \in X_{k+1} : A^k x \in X_\alpha\}$.

The family $(X_\alpha, A^\alpha)_{\alpha \in [-1, \infty)}$ is a densely injected Banach scale in the sense that $X_\alpha \xhookrightarrow{d} X_\beta$ whenever $\alpha \geq \beta$ and $A^\alpha : X_{\alpha+1} \rightarrow X_\alpha$ is an isomorphism for all $\alpha \in \mathbb{R}$. Moreover, $A^\alpha : X_{\alpha+1} \rightarrow X_\alpha$ is a densely defined closed operator with $0 \in \varrho(A^\alpha)$ for all $\alpha \in \mathbb{R}$. The family (X_α, A^α) is called an interpolation-extrapolation scale.

Theorem A.9. *Let A be the generator of a C_0 -semigroup $(S(t))_{t \geq 0}$ on X and let $\omega_S \in \mathbb{R}$ be the growth bound of the semigroup $S(t)$. Then $A^\alpha : X_{\alpha+1} \rightarrow X_\alpha$ also generates a C_0 -semigroup $(S_\alpha(t))_{t \geq 0}$ with the same growth bound. For all $\alpha, \beta \in [0, \infty)$ with $\alpha \geq \beta$ the following diagram commutes*

$$\begin{array}{ccc} X_\alpha & \xrightarrow{S_\alpha(t)} & X_\alpha \\ \downarrow & & \downarrow \\ X_\beta & \xrightarrow{S_\beta(t)} & X_\beta \end{array}$$

and for all $\omega > \omega_S$ there is a constant C depending on α and β such that

$$\|S_\beta(t)\|_{\mathcal{B}(X_\beta, X_\alpha)} \leq Ct^{\beta-\alpha} e^{\omega t} \quad \forall t > 0.$$

Appendix B. A different approach in controlling the fast-reaction term

In this section we present an alternative approach in constructing the slow manifold. We want to emphasize however that this is only a sketch of the approach and that the full development of this approach needs yet to be done.

The idea to the this approach comes from studying a finite dimensional fast-slow ODE. To this end, we study the following planar system

$$\frac{d}{dt} \begin{pmatrix} u^\varepsilon \\ v^\varepsilon \end{pmatrix} = \begin{pmatrix} -k^2 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} u^\varepsilon \\ v^\varepsilon \end{pmatrix} + \begin{pmatrix} \frac{1}{\varepsilon} f(u^\varepsilon, v^\varepsilon) \\ 0 \end{pmatrix}, \quad (\text{B.1})$$

where $k > 0$ is a fixed constant. In this example let $f(u, v) = -u + v^2$. Then, the ODE system has the explicit solution

$$u^\varepsilon(t) = u_0 e^{-t(k^2 + \varepsilon^{-1})} + \frac{(v_0)^2 e^{-2t}}{(-2 + k^2 + \varepsilon^{-1})\varepsilon}, \quad v^\varepsilon(t) = v_0 e^{-t}. \quad (\text{B.2})$$

The limit system as $\varepsilon \rightarrow 0$, which reads

$$\frac{d}{dt} v^0 = -v^0, \quad (\text{B.3})$$

defined on the critical manifold $S_0 = \{(u, v) \in \mathbb{R}^2 : u = v^2\}$, has the solution $v^0 = v_0 e^{-t}$.

Then, we can observe a different behavior of the solution in u for different ratios between k^2 and ε^{-1} . Assuming that $k^2 \ll \varepsilon^{-1}$ and the initial condition satisfy $u_0 = (v_0)^2$, then, the fast-reaction term is dominating and we obtain that $u^\varepsilon(t) = v^\varepsilon(t)^2 + \mathcal{O}(\varepsilon)$. This implies that in this regime for $t > 0$ we have

$$\begin{aligned} f(u^\varepsilon(t), v^\varepsilon(t)) &= (v_0)^2 \left(e^{-t(k^2 + \varepsilon^{-1})} - \frac{(-2\varepsilon + k^2\varepsilon)e^{-2t}}{-2\varepsilon + k^2\varepsilon + 1} \right) \\ &\leq \varepsilon C (v_0)^2 e^{-2t} = \varepsilon C h^0(v^0(t)), \end{aligned}$$

where $h^0(v) = v^2$. This estimate will be the key to compensate the ε^{-1} -term coming from the fast-reaction.

In the other regime, when $k^2 \geq \varepsilon^{-1}$, we observe that the linear (diffusion) term dominates and we have that $u^\varepsilon(t)$ decays to zero much faster than $v^\varepsilon(t)$ as t increases.

The next step is to generalize this idea to the PDE setting, where we consider a simplified fast-reaction equation of the form

$$\begin{aligned}\partial_t u^\varepsilon &= Au^\varepsilon + \frac{1}{\varepsilon} f(u^\varepsilon, v^\varepsilon), \\ u^\varepsilon(0) &= u_0,\end{aligned}\tag{B.4}$$

where $v^\varepsilon \in \mathbb{R}$ is a parameter depending on ε . The corresponding limit problem is

$$0 = f(u^0, v^0).\tag{B.5}$$

In order for this problem to be well-defined, we assume that f satisfies the assumptions of the implicit function theorem, such that there exists a function $h^0 : \mathbb{R} \rightarrow X_1$ with $f(h^0(v^0), v^0) = 0$. As before we further assume that the spectrum of $D_u f(u, v)$ is located on the left half-plane.

Motivated by the finite-dimensional ODE example we introduce a splitting of the Banach space X of the form

$$X = X_F^\xi \oplus X_S^\xi$$

into a fast part X_F^ξ , i.e. the case where the linear operator dominates the effect of the nonlinearity, and a slow part X_S^ξ , i.e. the case where the fast reaction term drives the flow of the solution. Here $\xi > 0$ is a small parameter encoding the splitting of the Banach space and we assume that the following holds

- The spaces X_F^ξ and X_S^ξ are closed in X and the projections $\text{pr}_{X_F^\xi}$ and $\text{pr}_{X_S^\xi}$ commute with A on X_1 .
- The spaces $X_S^\xi \cap X_1$ and $X_F^\xi \cap X_1$ are closed subspaces in X and will be endowed with the norm $\|\cdot\|_{X_1}$.
- The realization of the operator A in X_S^ξ is given by $A_{X_S^\xi} : D(X_S^\xi) \subset X_S^\xi \rightarrow X_S^\xi$ with

$$D(A_{X_S^\xi}) = \{u \in X_S^\xi \cap D(A) : Au \in X_S^\xi\}$$

and similar for the realization of A in X_F^ξ .

In addition, we make the following assumptions on the nonlinearity and the splitting based on the previous observations.

- The space X_F^ξ contains the parts of X_1 that decay very fast under the semigroup $(e^{tA})_{t \geq 0}$ satisfying the following estimate

$$\|e^{tA} x_F\|_{X_1} \leq M_A e^{\varepsilon^{-1} \omega_A t} \|x_F\|_{X_1},$$

where we observe a much faster decay when compared with the general estimate.

- The nonlinearity f satisfies

$$\|\text{pr}_{X_S^\varepsilon}[f(u_1, v_1) - f(u_2, v_2)]\|_{X_1} \leq \varepsilon L_f (\|u_1 - u_2\|_{X_1} + |v_1 - v_2|),$$

where $\text{pr}_{X_S^\varepsilon}$ denotes the projection onto X_S^ε and $u_1, u_2 \in X_1$ are solutions to (B.4) with parameters v_1 and v_2 respectively.

- We require the initial data under this splitting to satisfy

$$\text{pr}_{X_S^\varepsilon}[u_0 - h^0(v^0)] = 0.$$

Next, we show how to apply this method to equation (B.4), when studying the convergence of $u^\varepsilon \rightarrow h^0(v^0)$ in X_1 as $\varepsilon \rightarrow 0$. To simplify the computations, we assume that, as before in this section, the nonlinearity f is given by $f(u, v) = -u + v^2$. Then, setting $w^\varepsilon = u^\varepsilon - h^0(v^0)$ we obtain

$$\begin{aligned} \|w^\varepsilon(t)\|_{X_1} &= \left\| e^{At} w_0 + \varepsilon^{-1} \int_0^t e^{A(t-s)} f(w^\varepsilon(s) + h^0(v^0), v^\varepsilon) ds \right\|_{X_1} \\ &\leq M_A \int_0^t \left(e^{\omega_A(t-s)} + \varepsilon^{-1} e^{\varepsilon^{-1}\omega_A(t-s)} \right) (\|w^\varepsilon(s)\|_{X_1} + |h^0(v^0) - (v^\varepsilon)^2|) ds \\ &\quad + M_A e^{\varepsilon^{-1}\omega_A t} \|\text{pr}_{X_F^\varepsilon} w_0\|_{X_1}, \end{aligned}$$

where we used the splitting of the space X to control the ε^{-1} -terms. Noting that $h^0(v^0) = (v^0)^2$ we obtain

$$\begin{aligned} \|w^\varepsilon(t)\|_{X_1} &\leq M_A e^{\varepsilon^{-1}\omega_A t} \|\text{pr}_{X_F^\varepsilon} w_0\|_{X_1} + M_A |\omega_A^{-1}| |(v^0)^2 - (v^\varepsilon)^2| \\ &\quad + M_A \int_0^t \left(e^{\omega_A(t-s)} + \varepsilon^{-1} e^{\varepsilon^{-1}\omega_A(t-s)} \right) \|w^\varepsilon(s)\|_{X_1} ds. \end{aligned}$$

This then yields

$$\|w^\varepsilon(t)\|_{X_1} \leq C (e^{\varepsilon^{-1}\omega_A t} \|\text{pr}_{X_F^\varepsilon} w_0\|_{X_1} + |(v^0)^2 - (v^\varepsilon)^2|)$$

As the parameter $v^\varepsilon \rightarrow v^0$ as $\varepsilon \rightarrow 0$ we have $w^\varepsilon \rightarrow 0$, which shows the convergence of solutions.

Remark B.1. We want to point out that in this example we did not have to assume any global Lipschitz assumptions on the nonlinearity. However to compensate this lack of global control, more information on the specific problem at hand is required. This reason makes the application of this method to general problems more difficult and we leave this to future projects to investigate.

References

- [1] Amann, H.: Linear and quasilinear parabolic problems, vol. i. Birkhäuser, Basel (1995)

- [2] Bates, P.W., Lu, K., Zeng, C.: Existence and persistence of invariant manifolds for semiflows in banach space, vol. 645. American Mathematical Society, Providence RI (1998)
- [3] Bothe, D., Pierre, M.: The instantaneous limit for reaction-diffusion systems with a fast irreversible reaction. *Discrete and Continuous Dynamical Systems - S* **5**(1), 49–59 (2012)
- [4] Cantrell, R.S., Cosner, C.: On the dynamics of predator-prey models with the beddington-deangelis functional response. *J. Math. Anal. Appl.* **257**(1), 206–222 (2001)
- [5] Carrillo, J.A., Desvillettes, L., Fellner, K.: Fast-reaction limit for the inhomogeneous aizenman-bak model. *Kinetic & Related Models* **1**(1), 127 (2008)
- [6] Carrillo, J.A., Desvillettes, L., Fellner, K.: Rigorous derivation of a nonlinear diffusion equation as fast-reaction limit of a continuous coagulation-fragmentation model with diffusion. *Comm. Partial Differential Equations* **34**(11), 1338–1351 (2009)
- [7] Cesaroni, A., Dirr, N., Novaga, M.: Homogenization of a semilinear heat equation. *Journal de l'École polytechnique Mathématiques* **4**, 633–660 (2017)
- [8] Chow, S.-N., Lin, X.-B., Lu, K.: Smooth invariant foliations in infinite dimensional spaces. *J. Differential Equations* **94**(2), 266–291 (1991)
- [9] Chow, S.-N., Lu, K.: Invariant manifolds for flows in banach spaces. *J. Differential Equations* **74**(2), 285–317 (1988)
- [10] Conforto, F., Desvillettes, L., Soresina, C.: About reaction-diffusion systems involving the holling-type ii and the beddington-deangelis functional responses for predator-prey models. *Nonlinear Differ. Equ. Appl.* **25**(3), 24 (2018)
- [11] Crooks, E., Dancer, E., Hilhorst, D.: Fast reaction limit and long time behavior for a competition-diffusion system with dirichlet boundary conditions. *Discrete & Continuous Dynamical Systems-B* **8**(1), 39 (2007)
- [12] Daus, E., Desvillettes, L., Jüngel, A.: Cross-diffusion systems and fast-reaction limits. *Bulletin des Sciences Mathématiques* **159**, 102824 (2020)
- [13] De Masi, A., Funaki, T., Presutti, E., Vares, M.E.: Fast-reaction limit for glauher-kawasaki dynamics with two components. *Lat. Am. J. Probab. Math. Stat* **16**, 957–976 (2019)
- [14] Debussche, A., Temam, R.: Inertial manifolds and slow manifolds. *Appl. Math. Lett.* **4**(4), 73–76 (1991)
- [15] Disser, K., Liero, M., Zinsl, J.: Evolutionary γ -convergence of gradient systems modeling slow and fast chemical reactions. *Nonlinearity* **31**(8), 3689 (2018)
- [16] Eilertsen, J., Tyczynska, M.A., Schnell, S.: Hunting ε : The origin and validity of quasi-steady-state reductions in enzyme kinetics. *SIAM J. Appl. Dyn. Syst.* **20**(4), 2450–2481 (2021)

- [17] Engel, M., Hummel, F., Kuehn, C.: Connecting a direct and a galerkin approach to slow manifolds in infinite dimensions. *Proceedings of the American Mathematical Society, Series B* **8**(21), 252–266 (2021)
- [18] Engel, M., Hummel, F., Kuehn, C., Popović, N., Ptashnyk, M., Zacharis, T.: Geometric analysis of fast-slow PDEs with fold singularities via Galerkin discretisation. *Nonlinearity* **37**(11), 115017 (2024)
- [19] Engel, M., Kuehn, C.: *Blow-up analysis of fast-slow pdes with loss of hyperbolicity*, arXiv preprint [arXiv:2007.09973](https://arxiv.org/abs/2007.09973), 1–35 (2020)
- [20] Engel, M., Kuehn, C.: Geometric analysis of a truncated galerkin discretization of fast-slow PDEs with transcritical singularities. *SIAM J. Appl. Dyn. Syst.* **23**(4), 2853–2898 (2024)
- [21] Escobedo, M., Laurennot, P., Mischler, S.: Fast reaction limit of the discrete diffusive coagulation-fragmentation equation. *Comm. Partial Differential Equations* **28**(5–6), 1113–1133 (2003)
- [22] Fellner, K., Rosenberger, S., Tang, B.Q.: Quasi-steady-state approximation and numerical simulation for a volume-surface reaction-diffusion system. *Commun. Math. Sci.* **14**(6), 1553–1580 (2016)
- [23] Fenichel, N.: Persistence and smoothness of invariant manifolds for flows. *Indiana Univ. Math. J.* **21**(3), 193–226 (1971)
- [24] Fenichel, N.: Geometric singular perturbation theory for ordinary differential equations. *J. Differential Equations* **31**(1), 53–98 (1979)
- [25] Goeke, A., Schilli, C., Walcher, S., Zerz, E.: Computing quasi-steady state reductions. *J. Math. Chem.* **50**(6), 1495–1513 (2012)
- [26] Goutsias, J.: Quasiequilibrium approximation of fast reaction kinetics in stochastic biochemical systems. *J. Chem. Phys.* **122**(18), 184102 (2005)
- [27] Guarguaglini, F.R., Natalini, R.: Fast reaction limit and large time behavior of solutions to a nonlinear model of sulphation phenomena. *Comm. Partial Differential Equations* **32**(2), 163–189 (2007)
- [28] Haragus, M., Iooss, G.: *Local bifurcations, center manifolds, and normal forms in infinite-dimensional dynamical systems*, Springer, (2010)
- [29] Hausberg, S., Röger, M.: Well-posedness and fast-diffusion limit for a bulk-surface reaction-diffusion system. *Nonlinear Differ. Equ. Appl.* **25**(3), 1–32 (2018)
- [30] Henneke, F., Tang, B.Q.: Fast reaction limit of a volume-surface reaction-diffusion system towards a heat equation with dynamical boundary conditions. *Asymptot. Anal.* **98**(4), 325–339 (2016)
- [31] Henry, D.: *Geometric theory of semilinear parabolic equations*, vol. 840. Springer, Berlin (2006)

- [32] Herrmann, M., Niethammer, B., Velazquez, J.: Rate-independent dynamics and kramers-type phase transitions in nonlocal fokker-planck equations with dynamical control. *Arch. Ration. Mech. Anal.* **214**(3), 803–866 (2014)
- [33] Hilhorst, D., Van Der Hout, R., Peletier, L.: The fast reaction limit for a reaction-diffusion system. *J. Math. Anal. Appl.* **199**(2), 349–373 (1996)
- [34] Hummel, F., Jelbart, S., Kuehn, C.: Geometric blow-up of a dynamic Turing instability in the Swift-Hohenberg equation. *J. Differen. Equat.* **427**, 219–309 (2025)
- [35] Hummel, F., Kuehn, C.: Slow manifolds for infinite-dimensional evolution equations. *Commentarii Mathematici Helvetici* **97**(1), 61–132 (2022)
- [36] Iida, M., Ninomiya, H., Yamamoto, H.: A review on reaction-diffusion approximation. *Journal of Elliptic and Parabolic Equations* **4**(2), 565–600 (2018)
- [37] Jelbart, S., Kuehn, C.: A formal geometric blow-up method for pattern forming systems, *Topics in Multiple Time Scale Dynamics* (M. Engel, H. Jardón-Kojakhmetov, and C. Soresina, eds.), *Contemp. Math.*, vol. 806, Amer. Math. Soc., Providence, RI, 49–86 (2024)
- [38] Jelbart, S., Kuehn, C., Sánchez, A.M.: *Characterising exchange of stability in scalar reaction-diffusion equations via geometric blow-up*, [arXiv:2411.13679](https://arxiv.org/abs/2411.13679), 1–37 (2024)
- [39] Jelbart, Samuel: K Uldall Kristiansen, Peter Szmolyan, and Martin Wechselberger, Singularly perturbed oscillators with exponential nonlinearities. *J. Dyn. Diff. Equat.* **34**(3), 1823–1875 (2022)
- [40] Jones, C.: *Geometric singular perturbation theory*, *Dynamical systems*, 44–118 (1995)
- [41] Kostianko, A., Zelik, S.: *Kwak transform and inertial manifolds revisited*, *Journal of Dynamics and Differential Equations*, 1–21 (2021)
- [42] Kuehn, C.: *Multiple time scale dynamics*, vol. 191. Springer, Cham (2015)
- [43] Kuehn, C., Berglund, N., Bick, C., Engel, M., Hurth, T., Iuorio, A., Soresina, C.: A general view on double limits in differential equations. *Physica D* **431**, 133105 (2022)
- [44] Kuehn, C., Lehner, P., Sulzbach, J.-E.: *Infinite dimensional slow manifolds for a linear fast-reaction system*, *Topics in Multiple Time Scale Dynamics* (M. Engel, H. Jardón-Kojakhmetov, and C. Soresina, eds.), *Contemp. Math.*, vol. 806, Amer. Math. Soc., Providence, RI, 87–104 (2024)
- [45] Lunardi, A.: *Analytic semigroups and optimal regularity in parabolic problems*. Springer Science & Business Media, Basel (2012)
- [46] Mallet-Paret, J., Sell, G.: Inertial manifolds for reaction diffusion equations in higher space dimensions. *J. Am. Math. Soc.* **1**(4), 805–866 (1988)

- [47] Meier, S., Muntean, A.: *A two-scale reaction-diffusion system: homogenization and fast-reaction limits*, Current advances in nonlinear analysis and related topics, Gakkotosho, Tokyo, 441–459 (2010)
- [48] Mielke, A., Peletier, M., Stephan, A.: EDP-convergence for nonlinear fast-slow reaction systems with detailed balance. *Nonlinearity* **34**(8), 5762 (2021)
- [49] Murakawa, H., Ninomiya, H.: Fast reaction limit of a three-component reaction-diffusion system. *J. Math. Anal. Appl.* **379**(1), 150–170 (2011)
- [50] Richards, I.: On the gaps between numbers which are sums of two squares. *Adv. Math.* **46**(1), 1–2 (1982)
- [51] Skrzeczowski, J.: Fast reaction limit and forward-backward diffusion: A radon-nikodym approach. *Comptes Rendus. Mathématique* **360**(G2), 189–203 (2022)
- [52] Soresina, C., Tang, B.Q., Tran, B.-N.: *Fast-reaction limits for predator-prey reaction-diffusion systems: improved convergence*, Topics in Multiple Time Scale Dynamics, Contemp. Math., vol. 806, Amer. Math. Soc., Providence, RI, 173–188 (2024)
- [53] Stephan, A.: EDP-convergence for a linear reaction-diffusion system with fast reversible reaction. *Calc. Var. Partial. Differ. Equ.* **60**(6), 1–35 (2021)
- [54] Temam, R.: *Infinite-dimensional dynamical systems in mechanics and physics*, vol. 68. Springer Science & Business Media, New York (2012)
- [55] Vanderbauwhede, A., Iooss, G.: Center manifold theory in infinite dimensions, pp. 125–163. Springer, Dynamics Reported (1992)
- [56] Wechselberger, M.: *Geometric singular perturbation theory beyond the standard form*, vol. 6. Springer, New York (2020)
- [57] Zelik, S.: Inertial manifolds and finite-dimensional reduction for dissipative pdes. *Proceedings of the Royal Society of Edinburgh Section A: Mathematics* **144**(6), 1245–1327 (2014)

Christian Kuehn and Jan-Eric Sulzbach
Department of Mathematics
Technical University of Munich
Munich
Germany
e-mail: janeric.sulzbach@ma.tum.de

Christian Kuehn
e-mail: ckuehn@ma.tum.de

Received: 7 March 2024.

Revised: 11 March 2025.

Accepted: 8 May 2025.